

Tier-2 theory note: one geometry for PR-MPPI and CAMP-MPPI

Working note for the unified journal version

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Abstract

This note develops the theoretical backbone of the unified paper that merges PR-MPPI (velocity level, RA-L 26-4730) and CAMP-MPPI (acceleration level, ICRA 2027 #281). All new empirical results reported here are numerical studies or simulations; the timing results are computational benchmarks, and no physical-robot experiments are reported in this version. Both controllers project sampled inputs inside the rollouts and differ in order and metric. We place them in one geometry and prove seven results that neither manuscript contains: a separation theorem that says exactly when wrench and motion corrections interfere and why the inertia metric is the right one; an assignment of physical and virtual equalities to the order at which they must be enforced; an order-2 finite-step safety layer with a closed-form leading term; execution feasibility for both orders, by convexity after averaging or by a single final projection; a Proposition bounding the effect of off-manifold rollouts, restated at fixed horizon time; an exact identity for the distortion that inequality clipping causes in the MPPI update; and a finite-sample error bound for the update. The results assemble into one controller with per-cycle guarantees (Algorithm 1, Theorem 2). It takes the motion part of its command from the raw MPPI average, projected once, with the margin stated as a tightening of the rows by a calibrated execution-error bound, and its reaction from the average of the clamped sample reactions: averaging projected samples, as both earlier controllers do, keeps the motion off the binding rows by an amount nobody controls, while a raw average of the reaction passes the exploration noise along the contact normal into the contact force. A re-implementation of both controllers on the curved-surface writing task of CAMP-MPPI reproduces the force prediction of Proposition 2, the separation theorem, and the failure of a controller that treats a virtual equality as physical (Section 9). On the dual-arm task of PR-MPPI, a millimetre of base calibration error builds the internal wrench that the closed-chain form of the same proposition predicts, and anchoring or the order-2 controller removes it (Section 10). Four further propositions keep the certificate when the interaction model is uncertain. Rows multilinear in the parameters are made robust over a box by its vertices, a margin in the rows carries the certificate from the model wrench to the realized wrench (pointwise or on period averages, according to how the deviation is bounded), and set-membership adaptation shrinks the box online without losing the certificate (Section 11). On a waiter task with a tall box, a nominal model tips the box in every run at the main height, and no controller with certified rows ever does (0 of 61 runs); the simulation verifies the period-averaged form of the margin, not the pointwise one. With the force–torque sensor alone, centre-of-pressure differences close 77% of the tracking gap between the robust controller and an oracle, and all of it for lower centres of mass, with measurement and execution bounds calibrated in separate runs. A GPU implementation with one sample per thread and exact projections plans in 4.4ms for the writing task and 8.7ms for the waiter at $K = 1024$, inside their 10 and 20ms periods, and there the separable solver needs about half the kernel time of the joint projection (Section 12). Against tuned baselines (Section 13), penalty MPPI exceeded the true rows near the top of the prior, a one-step program without horizon tracked the waiter at 76.5mm against 26–39mm for Algorithm 1, and certainty equivalence came within 3mm of the oracle without violations but without a guarantee. Projecting the raw average once with tightened rows tracked 16% better than averaging projected samples while certifying the period-averaged realized wrench,

and at equal certificates it beat a penalty planner with a safety filter by 19%. Re-implementations of three safe-MPPI methods on the same model and rows show where the structure matters: GS-MPPI, whose raw average and closed-form filter match the motion part of Algorithm 1, matched it on the waiter but held the writing band only half of the time, DualGuard MPPI held the band but was the most conservative on the waiter, and Shield-MPPI trailed both. Where several support rows bind at once, GS-MPPI trailed Algorithm 1 by 3–5% in every seed, because its sequential half-space steps stop on one face of a vertex that the projection reaches, and its safe law applied once, as published, let the command reach 2.6 times the true slip limit. A robust offline plan after Heins and Schoellig tracked best but did not certify its executed commands (the box tipped at the top of the prior), and a robust NMPC with the same cost tracked 14–15% worse at about 0.45 s per step. On the dual-arm task, Shield-MPPI and DualGuard MPPI kept the joint bound closer to the goal than the calibrated acceleration margin of Algorithm 1 allows, which points to a position-level bound for joint rows. Every result except the one-line interpretation in Proposition 6(a) and the leading-order estimates of Proposition 4 is checked numerically by the scripts in `tier2/verify` (63 checks, all passing).

1 Summary

Three statements in the earlier plan are corrected here (Section 15). The support-wrench constraint of the CAMP tray task is a motion constraint, not a coupled one. The idea of a wrench-bounded retraction is replaced by the physical/virtual assignment of Proposition 2. The June off-manifold Proposition holds as stated at fixed horizon length; at fixed horizon time it needs the Δt -weighted cost of Proposition 5(A4), with which it becomes first order in Δt .

Table 1: Results of this note and the question each one answers.

Result	Question answered	Status	Script
Thm. 1	When does a wrench correction change the motion, and when is a joint projection necessary?	proved	v1
Prop. 1	Where do genuinely coupled constraints come from?	proved	v1
Prop. 2	Which equality is enforced at which order?	proved	v2
Lem. 1	How large a safety margin does the order-2 controller need?	proved	v3
Prop. 4	What is the leading term of that margin, and its sign?	leading order (sketch)	v3
Prop. 3	Is the executed command admissible, by convexity after averaging or by one final projection?	proved	v5, v9
Prop. 5	Rollouts leave the manifold; does it matter?	proved	v4
Prop. 6	Are both averages principled, and what does clipping distort?	proved	v5
Lem. 2	How accurate is the update with K samples?	proved	v6, v9
Prop. 7	How should an order-1 controller treat a physical row?	proved	writing
Prop. 8	What happens when the projection is infeasible?	proved	not triggered
Thm. 2	What does the unified controller guarantee per cycle?	proved	writing
Sec. 9	Do the predictions hold with re-implemented controllers on the CAMP task?	simulated	writing
Sec. 10	Do they hold on the PR-MPPI dual-arm task, with a calibration error?	simulated	dual-arm
Prop. 9	How do wrench rows stay exact when object parameters are known only to an interval?	proved	waiter
Props. 10, 12	Does the certificate survive, and can the interval shrink online without losing it?	proved	v8, waiter
Prop. 11	How does the certificate reach the realized wrench?	proved	v8, waiter
Sec. 11	Does this keep an uncertain object from tipping at speed, and what does adaptation buy?	simulated	waiter
Sec. 12	Is the controller real-time, and what does the separable solver save?	measured	GPU
Sec. 13	How does it compare with tuned baselines and safe-MPPI methods, also where several rows bind at once?	simulated	all tasks

2 Positioning against prior work

Four lines of work each cover part of what the unified controller does. Table 2 summarises them; the verdicts are for the constraint types listed, and “soft” means a penalty or slack.

Force–motion control. Hybrid position/force control [15] and the operational-space formulation [1] decouple motion and force, and Gauss’s principle gives the inertia-metric projection onto equalities that underlies them [2, 4]. Hierarchical QPs and task-space inverse dynamics add friction cones and limits [16, 17, 18], and CBF–QPs add safety rows [19, 20]. All of them solve one convex problem per instant: there is no horizon and no nonconvex cost. Theorem 1 extends the inertia-metric decoupling from equalities to inequality polyhedra and states when it fails.

Constrained and robust sampling-based MPC. MPPI [7] handles constraints as costs. Tube and robust MPPI [21, 22] bound the cost under disturbances, not constraint violation. Shield-MPPI [23] penalises the CBF condition in its rollouts and repairs the output by a fixed number of gradient steps, while composite-CBF MPPI [24] and DualGuard MPPI [25] make every sample safe, by a closed-form filter onto a soft-min composite barrier and by a Hamilton–Jacobi least-restrictive filter. None of the three handles equality constraints or interaction wrenches; Section 13 compares re-implementations of all three on the waiter task. Projection-based variants project samples in control space, onto input-derivative bounds [26] or task equalities with soft inequalities [27]. PR-MPPI [28] makes closed-chain equalities exact but has no wrench rows. Classical robust MPC tightens constraints over tubes for linear systems [29, 30].

Set-membership adaptive control. Adaptive MPC shrinks a parameter set online by set membership and keeps robust constraint satisfaction and recursive feasibility [31, 32, 33], including a nonlinear framework [34]. Robust adaptive CBFs do the same for a one-step safety filter [35]. In robotics, Ramdani and Poignet identify robot parameters offline with set-membership bounds [36]. Zhang et al. identify payload inertial parameters online with rigorous bounds under bounded sensor noise, along dedicated exciting trajectories that are planned to be safe [37]. In both, identification is a stage with its own trajectories, and neither handles the support wrench of a non-prehensile object. Here the set is updated during the task itself, from the measurements the task produces, inside the wrench constraints that the set certifies. None of these methods samples trajectories.

Non-prehensile transport. Waiter MPC with friction-cone constraints is gradient-based and softens its constraints [38, 41]; see [42] for a survey. The closest work is [39]. It enforces the sticking constraints at every vertex of a fixed polytope of centre-of-mass positions, which is the vertex reduction of Proposition 9, but it plans once offline, softens the constraints, and never updates the set. Serving Time [40] certifies contact-wrench constraints under small fixed intervals by reachability analysis, and does no estimation.

What is new. Enforcing hard wrench and HOCBF rows on every MPPI sample in the inertia metric, with feasibility of the executed command at $t = 0$, is already CAMP-MPPI, and exact equalities on every sample at order 1 are PR-MPPI. Relative to both manuscripts and to the work above, the note contributes three connected results.

1. *The condition for separation* (Theorem 1). It states exactly which metrics decouple wrench and motion corrections under inequality rows: $\text{range}(W^{-1}J_c^\top) = \text{range}(M^{-1}J_c^\top)$, which holds for the inertia metric and the family $M + J_c^\top S J_c$. The separable solver and the unconditional admissibility of the reactions (Proposition 8) follow from it. CAMP-MPPI chooses the inertia metric without this characterisation and solves one composite projection; the metric itself is classical for equalities [1, 4].
2. *Physical versus virtual equalities* (Propositions 2 and 7). These results say which equality is enforced at which order: the order-2 controller is invariant to surface offsets, the order-1 controller

Table 2: Capabilities of the closest prior work. “1-step” means a single convex problem per control instant; [†] marks the methods re-implemented and compared in Section 13.

Work	Sampling MPC, nonconvex cost	Hard wrench rows	Certified executed command	Uncertain object parameters	Online set shrinking
OSC, Gauss [1, 4]	–	equalities	1-step	–	–
HQP, TSID [17, 18]	–	✓	1-step	–	–
CBF-QP, HOCBF [19, 20]	–	–	1-step	–	–
MPPI, tube/robust MPPI [7, 22]	✓	–	–	disturbances	–
Shield-MPPI [†] [23]	✓	–	local repair	–	–
GS-MPPI [†] [24]	✓	–	state barriers	–	–
DualGuard MPPI [†] [25]	✓	–	HJ safe set	disturbances	–
PR-MPPI [28]	✓	–	equalities, motion rows	–	–
CAMP-MPPI (under review)	✓	✓	at $t = 0$	–	–
Set-membership adaptive MPC [32, 34]	–	–	✓	✓	✓
Robust adaptive CBF [35]	–	–	1-step	✓	✓
Robust waiter planning [†] [39]	–	soft	–	vertices, fixed	–
Serving Time [40]	–	✓	✓	small, fixed	–
This note	✓	✓	✓	vertices	✓

has a closed-form force error, and anchoring removes it. CAMP-MPPI keeps the pen orientation as a cost; as virtual rows it needs no weight and gives a sixth of the tilt, while placing it inside the manifold fails as predicted (Section 9).

3. *Certified wrench rows under model uncertainty* (Propositions 9–12). Robust rows over a parameter box keep the certificate, a row margin carries it from the model wrench to the realized wrench (pointwise or on period averages), and set-membership estimation shrinks the box during the task, inside the wrench constraints it certifies, without a separate identification stage. The margin replaces the one that both earlier controllers leave implicit: they execute the average of the projected samples, which the clipping distortion keeps off the binding rows (Proposition 6(b)), whereas Algorithm 1 projects the raw average of the motion once, keeps the average of the clamped samples for the reaction, and tightens the rows by a calibrated execution-error bound, which tracked better and certifies the realized wrench (Section 13). This split, which separation allows, is also what lets one controller do well both where motion rows bind and where a contact force must be held: of the re-implemented Shield-MPPI, DualGuard MPPI and GS-MPPI, none does both (Section 13). Where several motion rows bind at once, the exact projection also beats the closed-form filter of GS-MPPI, which applied once leaves input rows violated and applied repeatedly stops on one face of a vertex. Vertex enumeration over a fixed centre-of-mass set is known [39], and guaranteed identification along dedicated exciting trajectories is known [37]; what is new is the update during the task with the certificate kept.

The per-cycle guarantees of Theorem 2 assemble these results into one algorithm for both orders; the finite-step layer, the clipping identity and the estimation bound support it.

Overlap with the two manuscripts. T-RO requires “new results of substantive research significance and impact beyond the previous papers” for submissions based on earlier publications, and it does not consider work that is partly under review elsewhere. Because RA-L is an archival journal,

the RA-L FAQ asks that a later Transactions paper “have little overlap with a paper published in the Letters, and contain mainly novel material”. The unified paper should therefore cite PR-MPPI as the order-1 member of the family and not re-derive it. Its body should be the material of this note: Theorems 1 and 2, Propositions 2–8, and Section 11. It can be submitted only after both manuscripts are decided.

3 Setting

Let $q \in \mathbb{R}^n$ with inertia $M(q) \succ 0$ and bias $b(q, \dot{q})$. An equality $c : \mathbb{R}^n \rightarrow \mathbb{R}^{n_e}$ has Jacobian $J_c = \partial c / \partial q$ of full row rank on the operating set, with $\sigma_{\min}(J_c) \geq \underline{\sigma} > 0$. We write $\mathcal{M} = \{c = 0\}$ and $T\mathcal{M} = \{(q, \dot{q}) : c(q) = 0, J_c \dot{q} = 0\}$.

For a metric $W \succ 0$ the W -weighted pseudoinverse and projector are

$$J_c^{\#W} = W^{-1} J_c^\top (J_c W^{-1} J_c^\top)^{-1}, \quad P_W = I - J_c^{\#W} J_c. \quad (1)$$

P_W is the W -orthogonal projector onto $\mathcal{T} := \ker J_c$. PR-MPPI uses $N := P_I$. CAMP-MPPI uses $F := P_M = I - M^{-1} J_c^\top \Lambda_c J_c$ with $\Lambda_c = (J_c M^{-1} J_c^\top)^{-1}$, which is the dynamically consistent null-space projector of Khatib [1].

Order 1 (PR-MPPI). The input is $u = \dot{q}$, the rollout is $q_{t+1} = q_t + \Delta t \Pi^{(1)}(q_t, \tilde{u}_t)$, and $\Pi^{(1)}(q, \tilde{u}) = \arg \min_u \frac{1}{2} \|u - \tilde{u}\|_W^2$ subject to $J_c u = 0$ and $J_h u \geq -\Gamma \bar{h}$.

Order 2 (CAMP-MPPI). The input is a commanded acceleration a applied through $\tau = Ma + b$. Gauss’ principle [2, 3] gives the realised acceleration and the reaction

$$\ddot{q}(a) = \arg \min_{x: J_c x = -J_c \dot{q}} \frac{1}{2} \|x - a\|_M^2 = Fa + f, \quad \lambda(a) = -\Lambda_c (J_c a + \dot{J}_c \dot{q}), \quad (2)$$

with $f = -J_c^{\#M} \dot{J}_c \dot{q}$. This is exactly the constrained dynamics of CAMP-MPPI, Eqs. (15), (18) and (30)–(31). Both controllers are therefore projections in a metric; they differ in the order of the projected quantity and in the choice $W = I$ versus $W = M$. Since λ denotes the reaction, the MPPI temperature is written β , as in CAMP-MPPI.

4 Constraint classes and the separation theorem

Fix a state $(q, \dot{q})$ and let $\mathcal{N} := \text{range}(M^{-1} J_c^\top)$. All constraints of the order-2 controller are affine rows $c_i^\top a \leq d_i$ in the commanded acceleration.

Definition 1 (Constraint classes). A row is a *motion row* if $c_i^\top M^{-1} J_c^\top = 0$, a *reaction row* if $c_i \in \text{range}(J_c^\top)$, and a *coupled row* otherwise.

A row is a motion row if and only if it is an affine constraint on the realised acceleration $\ddot{q}(a)$, and a reaction row if and only if it is an affine constraint on the reaction $\lambda(a)$. Indeed $c_i^\top (I - F) = c_i^\top M^{-1} J_c^\top \Lambda_c J_c$ vanishes for motion rows, so $c_i^\top a = c_i^\top (\ddot{q}(a) - f)$, while $c_i = J_c^\top y_i$ gives $c_i^\top a = -y_i^\top (\Lambda_c^{-1} \lambda(a) + \dot{J}_c \dot{q})$. HOCBF rows (CAMP Eq. (32)) and joint-limit rows are motion rows. Bounds on the contact force of a physical manifold (CAMP Eq. (26) with $G = -\Lambda_c J_c$) are reaction rows. Torque limits $|Ma + b| \leq \bar{\tau}$ are coupled rows.

Theorem 1 (Separation). Let $\Phi(a) := (\ddot{q}(a), \lambda(a))$.

(i) $\mathbb{R}^n = \mathcal{T} \oplus \mathcal{N}$ with $\mathcal{T} \perp_M \mathcal{N}$, and Φ is an affine bijection from $\mathbb{R}^n$ onto $(f + \mathcal{T}) \times \mathbb{R}^{n_e}$ with inverse $\Phi^{-1}(\ddot{q}, \lambda) = (\ddot{q} - f) - M^{-1}J_c^\top(\lambda + \Lambda_c \dot{J}_c \dot{q})$.

(ii) For all a, a' ,

$$\|a - a'\|_M^2 = \|\ddot{q}(a) - \ddot{q}(a')\|_M^2 + \|\lambda(a) - \lambda(a')\|_{\Lambda_c^{-1}}^2. \quad (3)$$

(iii) Let $\mathcal{A}$ be cut out by motion rows and reaction rows only, let $\mathcal{Q} \subset f + \mathcal{T}$ be the admissible accelerations and $\Omega \subset \mathbb{R}^{n_e}$ the admissible reactions, both closed and convex. Then $\mathcal{A} = \Phi^{-1}(\mathcal{Q} \times \Omega)$, and if $\mathcal{A} \neq \emptyset$ the M -projection a^* of any $\tilde{a}$ onto $\mathcal{A}$ satisfies

$$\ddot{q}(a^*) = \arg \min_{x \in \mathcal{Q}} \|x - \ddot{q}(\tilde{a})\|_M, \quad \lambda(a^*) = \arg \min_{y \in \Omega} \|y - \lambda(\tilde{a})\|_{\Lambda_c^{-1}}. \quad (4)$$

In particular a correction triggered only by reaction rows leaves the motion unchanged, and a correction triggered only by motion rows leaves the reaction unchanged.

(iv) For a metric $W \succ 0$, the W -projection onto $\mathcal{A}$ decouples for every $\tilde{a}$ and every choice of motion and reaction rows (the motion of a^* depends only on $\ddot{q}(\tilde{a})$ and the motion rows, its reaction only on $\lambda(\tilde{a})$ and the reaction rows, each in the metric that W induces on $\mathcal{T}$ or $\mathcal{N}$) if and only if $\text{range}(W^{-1}J_c^\top) = \mathcal{N}$, that is, if and only if $\mathcal{T} \perp_W \mathcal{N}$. This holds for $W = M$ and for $W = M + J_c^\top S J_c$ with $S \succeq 0$, and fails for $W = I$ unless $\text{range}(J_c^\top)$ is M -invariant.

Proof. (i) For $x \in \mathcal{T}$ and $y = M^{-1}J_c^\top z \in \mathcal{N}$, $\langle x, y \rangle_M = (J_c x)^\top z = 0$. If $M^{-1}J_c^\top z \in \mathcal{T}$ then $\Lambda_c^{-1}z = 0$, so $\mathcal{T} \cap \mathcal{N} = \{0\}$ and the dimensions add to n . F is the projector onto $\mathcal{T}$ along $\mathcal{N}$, so $\ddot{q}(a) = Fa + f$ depends only on the $\mathcal{T}$ -component of a . Since $J_c a$ depends only on the $\mathcal{N}$ -component, so does $\lambda(a)$. For $a_{\mathcal{N}} = M^{-1}J_c^\top z$ we get $\lambda = -z - \Lambda_c \dot{J}_c \dot{q}$, an affine bijection $\mathcal{N} \rightarrow \mathbb{R}^{n_e}$, which gives the inverse formula.

(ii) Split $a - a'$ into its M -orthogonal components. The $\mathcal{T}$ -component equals $\ddot{q}(a) - \ddot{q}(a')$, and the $\mathcal{N}$ -component $M^{-1}J_c^\top \Delta z$ has $\|\cdot\|_M^2 = \Delta z^\top \Lambda_c^{-1} \Delta z$ with $\Delta z = -\Delta \lambda$.

(iii) Motion rows constrain $\ddot{q}$ only and reaction rows constrain λ only, so $\mathcal{A} = \Phi^{-1}(\mathcal{Q} \times \Omega)$. By (ii) the projection objective in the coordinates $(\ddot{q}, \lambda)$ is a sum of a term in $\ddot{q}$ and a term in λ , over a product set.

(iv) If $\mathcal{T} \perp_W \mathcal{N}$, the objective $\|a - \tilde{a}\|_W^2$ splits over the same decomposition while motion rows act on the $\mathcal{T}$ -component and reaction rows on the $\mathcal{N}$ -component, so the problem separates. $\mathcal{T} \perp_W \mathcal{N}$ is equivalent to $W\mathcal{N} \subseteq \mathcal{T}^\perp = \text{range}(J_c^\top)$, i.e. to $\mathcal{N} = \text{range}(W^{-1}J_c^\top)$ by equal dimensions. Conversely, if $\text{range}(W^{-1}J_c^\top) \neq \mathcal{N}$ there is y with $W^{-1}J_c^\top y \notin \mathcal{N} = \ker F$. Take the single reaction row $(J_c^\top y)^\top a \leq d$ with d violated by $\tilde{a}$. The W -projection is $\tilde{a} - \mu W^{-1}J_c^\top y$ with $\mu > 0$ and changes the motion by $-\mu F W^{-1}J_c^\top y \neq 0$. For $W = M + J_c^\top S J_c$ and $x \in \mathcal{T}$, $x^\top W M^{-1}J_c^\top z = (J_c x)^\top (z + S J_c M^{-1}J_c^\top z) = 0$. $\square$

Corollary 1 (Reaction rows never cost motion). *Under the assumptions of Theorem 1(iii), $\mathcal{A} \neq \emptyset$ if and only if $\mathcal{Q} \neq \emptyset$ and $\Omega \neq \emptyset$. If $\Omega \neq \emptyset$, then $\ddot{q}(\mathcal{A}) = \mathcal{Q}$: reaction rows that are feasible among themselves can neither cause infeasibility nor restrict the achievable motion.*

Remark 1 (Why the inertia metric). Identity (3) is the answer that CAMP-MPPI Eq. (36) only half gives. The inertia metric measures a correction as the kinetic-metric change of the actual motion plus the inverse-effective-mass change of the reaction, with no cross term. Under $W = I$ a force correction leaks into the motion and vice versa; in v1 the leakage is up to 8×10^1 under $W = I$ and below 10^{-11} under $W = M$. Coordinate invariance is a second, weaker argument: P_M transforms covariantly under $q = Sq'$, P_I does not. It matters only for mixed units (mobile bases, floating bases), not for the all-revolute arms used so far. The family $W = M + J_c^\top S J_c$ keeps separation and changes only the reaction metric to $\Lambda_c^{-1} + \Lambda_c^{-1} S \Lambda_c^{-1}$, which is a free weighting knob for wrench corrections.

Remark 2 (Relation to operational-space control). Force–motion decoupling under the inertia metric is classical in operational-space and hybrid force/motion control [1, 5, 6], and identity (3) is its energetic form. The new elements are the characterisation of all separating metrics in (iv), Corollary 1, the source of coupling in Proposition 1, and their consequence for sampled projections. A sampled wrench correction never perturbs the rollout motion, and the composite projection of CAMP-MPPI splits into two smaller problems whenever no coupled row is present.

Remark 3 (Consequences for the two manuscripts). In the CAMP writing task the rows are motion rows (HOCBF) and reaction rows (normal-force band), so the composite projection splits into a tangential QP and an n_e -dimensional reaction QP; for a single normal force the latter is a clamp. In the CAMP tray task there is no manifold, and the support wrench is an affine function of the tray acceleration, hence of $\ddot{q}$; its rows are motion rows at order 2. The joint projection matters there because several motion rows interact, and it is unavailable at order 1 because the rows depend on acceleration.

Proposition 1 (Passive bodies create coupling). *If n_p coordinates are unactuated, the realisable commanded accelerations form an affine subspace $\mathcal{U} = -M^{-1}b + M^{-1}S^\top \mathbb{R}^{n-n_p}$, where S selects the actuated coordinates. Then $\Phi(\mathcal{U})$ is an affine subspace of dimension $n - n_p$ in the n -dimensional space $(f + \mathcal{T}) \times \mathbb{R}^{n_e}$, and reaction rows restrict the admissible motions in general.*

Proof. Φ is an affine bijection by Theorem 1(i), so it maps $\mathcal{U}$ onto an affine subspace of the same dimension. A product set $\mathcal{Q} \times \Omega$ intersected with a proper affine subspace is not a product in general. $\square$

Remark 4. This is the mechanism behind friction limits of a passive object squeezed by two arms. The object’s Newton–Euler equations tie its acceleration to the grasp reactions, so friction-cone rows on the reactions bound the motion. In `v1`, a reaction band leaves a motion LP unbounded in all 50 fully actuated trials and bounds it in all 50 trials with one passive coordinate. Passive bodies and torque limits are two sources of genuinely coupled rows, where the joint projection of CAMP-MPPI is necessary rather than convenient. They are not the only ones: any row that mixes the realised acceleration and the reaction is coupled, for instance a bound on the wrench at a wrist sensor that carries a tool mass.

5 Physical versus virtual equalities

Definition 2. An equality is *physical* if the environment or the mechanism supplies the reaction that enforces it (surface contact, a closed chain through a grasped object), and *virtual* if only the controller enforces it (tray levelness, tool orientation).

Proposition 2 (Order assignment).

- (a) *At order 2 the commanded reaction $\lambda(a)$ and the realised motion $\ddot{q}(a)$ depend on a physical equality only through J_c and $\dot{J}_c \dot{q}$ at the measured state. A model error that offsets c by a constant δ leaves both unchanged.*
- (b) *At order 1, let the controller command $q^+ \in \{c = 0\}$ for a physical equality whose true manifold is $\{c = \delta\}$, tracked by a joint-space loop of stiffness $K \succ 0$ with gravity compensation, against an environment of stiffness K_{env} in constraint coordinates. Let the contact be bilateral, or unilateral and closed, and let q^+ be held until static equilibrium. Then,*

$$\lambda = (\Lambda_K^{-1} + K_{\text{env}}^{-1})^{-1} \delta + O(\|\delta\|^2), \quad \Lambda_K = (J_c K^{-1} J_c^\top)^{-1}. \quad (5)$$

Proof. (a) Eq. (2) contains c only through J_c and $\dot{J}_c$, which a constant offset does not change. (b) Write $q = q^+ + \Delta$. Equilibrium of the loop gives $K\Delta = J_c^\top \lambda$. The contact gives $c(q) = \delta - K_{\text{env}}^{-1} \lambda$, and $c(q) = J_c \Delta + O(\|\Delta\|^2)$ because $c(q^+) = 0$. Hence $\Lambda_K^{-1} \lambda = \delta - K_{\text{env}}^{-1} \lambda + O(\|\delta\|^2)$. $\square$

Remark 5 (Sign, reference restarts and closed chains). With the sign of (5), a unilateral contact is pressed for one sign of δ and opens for the other, where $\lambda = 0$. In the writing experiment $\delta < 0$ denotes a board model behind the real board, which presses, and the tables report the magnitude. Two variants of K appear in the experiments. First, an order-1 executor that restarts an interpolated reference $q_{\text{ref}}(\tau) = q_0 + \tau u$ at the measured state every period, with gains $M\omega^2$ and $2M\omega$ and the step $u = (q^+ - q_0)/\Delta t$, holds a static contact with the period-averaged force $M(\omega^2 \Delta t/2 + 2\omega)u$. Its stiffness is therefore $K_{\text{eff}} = M(\omega^2/2 + 2\omega/\Delta t)$ instead of $M\omega^2$. Second, for a closed chain through a grasped object, c is the relative grasp pose, $J_c = [J_1, -J_2]$, $K = \text{diag}(K_1, K_2)$ stacks the two arms' loops, and K_{env} is the grasp stiffness, so (5) gives the internal wrench caused by a calibration offset δ of one base.

Corollary 2 (Design rule). *Virtual equalities are retracted: by the Gauss–Newton retraction at order 1, and by a phase-space retraction of the commanded reference at order 2. Physical equalities enter the Gauss dynamics with reaction rows at order 2. A position-level retraction onto a physical manifold turns geometric model error into reaction error at the series stiffness $(\Lambda_K^{-1} + K_{\text{env}}^{-1})^{-1}$, which no inequality of the order-1 controller can see.*

Remark 6 (Numbers and the two manuscripts). For the planar 3R arm of v2 the gain in (5) is 1.29 N/mm under a joint PD loop and 7.25 N/mm under a computed-torque loop of 10 Hz bandwidth; the first-order formula is accurate to within 5×10^{-4} at 0.05 mm offset. With the computed-torque loop, a 2–4 N contact band confines the surface model to a 0.28 mm window. Section 9 tests this mechanism in simulation by sweeping the board-model offset and comparing the measured contact force with Proposition 2(b). The PR-MPPI equality $c \in \mathbb{R}^8$ splits into six physical rows (relative gripper pose, enforced by the tray) and two virtual rows (tray roll and pitch). In the unified controller the six enter the Gauss dynamics with grasp-wrench rows and the two are retracted.

6 Order-unified guarantees

6.1 Execution feasibility

Proposition 3 (Feasibility at $t = 0$). *Let $\mathcal{A}_0$ be the admissible set at the measured state, a polyhedron intersected with the equality, and let Π_0 be the W -projection onto it.*

- (a) *At order 2 with $W = M$ and no coupled rows, let x be the M -projection of $F\hat{v}$ onto the motion rows for a raw average $\hat{v}$, and let $\bar{\lambda} \in \Omega$. Then $\Phi^{-1}(x + f, \bar{\lambda}) \in \mathcal{A}_0$ whenever the motion rows are feasible. This covers $\Pi_0(\hat{v})$, for which $\bar{\lambda}$ is the projection of $\lambda(\hat{v})$ onto Ω , and the average of admissible sample reactions, which lies in Ω because Ω is convex; at order 1, $\Pi_0(\hat{v}) \in \mathcal{A}_0$ whenever $\mathcal{A}_0 \neq \emptyset$. If every row $L_i y \leq b_i$ of $\mathcal{A}_0$ is tightened to $L_i y \leq b_i - \varepsilon_i \|L_i\|_*$, each untightened row holds for every realisation whose argument y of that row deviates from that of $\Pi_0(\hat{v})$ by at most ε_i .*
- (b) *If every sample's first applied input lies in $\mathcal{A}_0$, their weighted average lies in $\mathcal{A}_0$ and Π_0 acts on it as the identity. At order 2 the realised motion and reaction of that average are the averages of the samples' motions and reactions, because $\ddot{q}(\cdot)$ and $\lambda(\cdot)$ are affine.*

Proof. (a) By Theorem 1(i) the motion rows constrain only x and the reaction rows only λ , so $\mathcal{A}_0$ is the image under Φ^{-1} of a product, and a projection maps into its target set; at order 1 the

second statement is the same fact. Hölder's inequality gives $L_i y' \leq L_i y + \|L_i\|_* \|y' - y\| \leq b_i$, as in Proposition 11. (b) All samples share the state at $t = 0$, $\mathcal{A}_0$ is convex, and projections fix points of their target set. $\square$

Remark 7. Item (b) is the route of the two earlier controllers. In Full PR-MPPI the execution projection (Eq. (13)) is inactive at $t = 0$ and acts only on the warm start ($t \geq 1$) and in the execution-only ablation, and CAMP-MPPI Eqs. (49)–(54) are item (b) at order 2. Algorithm 1 uses item (a) for the motion part and keeps the convexity of item (b) for the reaction. By identity (9), the average of the projected samples differs from $\Pi_0(\hat{v})$ exactly by the clipping distortion of Proposition 6(b), which moves it off the active face into $\mathcal{A}_0$. That displacement acts as a margin whose size nobody controls; item (a) removes it from the motion rows and puts an explicit margin ε in its place (Section 13).

6.2 Finite-step safety layer

Lemma 1 of PR-MPPI covers order 1. The order-2 counterpart for the HOCBF of CAMP-MPPI Eq. (28) is new.

Lemma 1 (Order-2 finite-step layer). *Let $\dot{q}_{t+1} = \dot{q}_t + \Delta t \ddot{q}_t$ and $q_{t+1} = q_t + \Delta t \dot{q}_{t+1}$, where the realised $\ddot{q}_t$ satisfies $\ddot{h}_t + (k_1 + k_2)\dot{h}_t + k_1 k_2 h_t \geq 0$ with $\dot{h}_t = \nabla h^\top \dot{q}_t$ and $\ddot{h}_t = \nabla h^\top \ddot{q}_t + \dot{q}_t^\top \nabla^2 h \dot{q}_t$ at q_t . Assume $\|\nabla^j h\| \leq L_j$ ($j = 1, 2, 3$), $\|\dot{q}_t\| \leq \bar{v}$, $\|\ddot{q}_t\| \leq \bar{a}$ and $k_i \Delta t \leq 1$. With $\psi_t = \dot{h}_t + k_1 h_t$,*

$$\psi_{t+1} \geq (1 - k_2 \Delta t) \psi_t - \kappa_\psi \Delta t^2, \quad h_{t+1} \geq (1 - k_1 \Delta t) h_t + \Delta t \psi_t - \kappa_h \Delta t^2, \quad (6)$$

where $\kappa_h = \frac{1}{2} L_2 \bar{v}^2 + L_1 \bar{a}$ and $\kappa_\psi = 2 L_2 \bar{a} \bar{v} + L_2 \bar{a}^2 \Delta t + \frac{1}{2} L_3 \bar{v}^3 + k_1 \kappa_h$. If $\psi_0 \geq -\kappa_\psi \Delta t / k_2$ and $h_0 \geq -\underline{h}$ with $\underline{h} = (\kappa_h + \kappa_\psi / k_2) \Delta t / k_1$, then $h_t \geq -\underline{h}$ for all t . With fixed per-step gains $\bar{k}_i = k_i \Delta t$ the layer is $O(\Delta t^2)$.

Proof. Taylor expansion of h along the step gives $h_{t+1} = h_t + \Delta t \dot{h}_t + E_h$ with $E_h = \Delta t^2 \nabla h^\top \ddot{q}_t + \frac{1}{2} \Delta t^2 \dot{q}_{t+1}^\top \nabla^2 h(\xi) \dot{q}_{t+1}$, so $|E_h| \leq \kappa_h \Delta t^2$. Expanding $\nabla h(q_{t+1})^\top \dot{q}_{t+1}$ gives $\dot{h}_{t+1} = \dot{h}_t + \Delta t \ddot{h}_t + E_v$ with $|E_v| \leq (2 L_2 \bar{a} \bar{v} + L_2 \bar{a}^2 \Delta t + \frac{1}{2} L_3 \bar{v}^3) \Delta t^2$. The HOCBF condition is $\dot{h}_t + k_1 \dot{h}_t \geq -k_2 \psi_t$, which yields both recursions. The layers $-\kappa_\psi \Delta t / k_2$ and $-\underline{h}$ are fixed points of the recursions and are invariant because $1 - k_i \Delta t \geq 0$. $\square$

Remark 8 (Relation to sampled-data CBFs). Safety margins for discrete-time and zero-order-hold implementations of CBFs are known [11, 12, 13]. Lemma 1 specialises them to the semi-implicit rollout model of CAMP-MPPI. Proposition 4 adds what makes the margin selectable before deployment: the closed-form leading term and the order-dependent sign.

The bound is conservative (ratio 0.09 in v3). Under steady sliding along the boundary its leading term has a closed form, and the sign of the discretisation error depends on the order.

Proposition 4 (Leading-order layers). *Let h be a signed distance near the boundary ($\|\nabla h\| = 1$), and let the controller slide along a boundary of curvature radius ρ at tangential speed v .*

- (a) *Order 1, Euler, CBF gain γ : the one-step error is $+\frac{1}{2} \Delta t^2 u^\top \nabla^2 h u$, adverse where h is concave along the motion (inside a convex safe set), and the steady layer is $h^* = -v^2 \Delta t / (2\rho\gamma) + o(\Delta t)$.*
- (b) *Order 2, semi-implicit Euler, HOCBF gains k_1, k_2 : the one-step error is $\Delta t^2 (\ddot{h}_t - \frac{1}{2} \dot{q}^\top \nabla^2 h \dot{q})$, adverse where h is convex along the motion (outside a convex obstacle), and the steady layer is*

$$h^* = -\frac{v^2 \Delta t}{2\rho} \left(\frac{1}{k_1} + \frac{1}{k_2} \right) + o(\Delta t). \quad (7)$$

- (c) Order 2, zero-order-hold execution on the continuous system (the command held over Δt): the one-step errors are $E_h = \frac{1}{2}\Delta t^2 \ddot{h}_t + O(\Delta t^3)$ and $E_v = \frac{1}{2}\Delta t^2 \ddot{\dot{h}}_t + O(\Delta t^3)$, where $\ddot{h}_t$ is the barrier jerk under the held command, and the quasi-steady layer is

$$h^* = \frac{\Delta t \ddot{h}}{2k_1 k_2} + O(\Delta t^2). \quad (8)$$

In (a) and (b) the sign flips on the opposite curvature and the controller keeps a positive margin of the same size. In (c) the layer is adverse when $\ddot{h} < 0$, for instance when braking along a boundary, and it is of higher order during steady sliding, where $\ddot{h} \rightarrow 0$.

Proof sketch. (b) At steady sliding the HOCBF is active, $\dot{h} = O(\Delta t)$ and the tangential acceleration vanishes, which gives $E_v = o(\Delta t^2)$ and $E_h = -\frac{1}{2}\Delta t^2 v^2/\rho$. The fixed point of $h_{t+1} = h_t + \Delta t \dot{h}_t + E_h$ and $\dot{h}_{t+1} = \dot{h}_t - \Delta t(k_1 + k_2)\dot{h}_t - \Delta t k_1 k_2 h_t + E_v$ is $h^* = E_v/(k_1 k_2 \Delta t) + E_h(1/k_1 + 1/k_2)/\Delta t$. (a) is the same computation with a single recursion. (c) With the exact Taylor errors, the fixed point satisfies $\ddot{h}^* = -\frac{1}{2}\Delta t \ddot{\dot{h}}$ and $\dot{h}^* = \frac{1}{12}\Delta t^2 \ddot{\dot{h}}$, and the active HOCBF $\ddot{h} = -(k_1 + k_2)\dot{h} - k_1 k_2 h$ then gives (8). $\square$

Remark 9 (Margin rules). Lemma 1 is the only rigorous margin: $h_{\text{safe}} \geq \underline{h}$ keeps the barriers of the rollout model nonnegative under its hypotheses. Proposition 4 gives leading-order estimates, which are sharper but are not bounds. Part (b) suggests the rollout margin $h_{\text{safe}} \approx v_{\text{max}}^2 \Delta t (1/k_1 + 1/k_2)/(2\rho_{\text{min}})$, and part (c) the executed-motion margin $h_{\text{safe}} \approx \Delta t \max(-\ddot{h})/(2k_1 k_2)$ for an exact model, which replaces the generic control-interval margin of CAMP-MPPI with a bound on the barrier jerk. In v3 the measured layers match the closed forms to within 0.2% (a), 2% (b) and 6% (c). With fixed per-step gains the remainder of (c), $-\frac{1}{12}\Delta t^2 \ddot{\dot{h}}(1/k_1 + 1/k_2)$, has the same order as the leading term. It then predicts the order of the layer, $O(\Delta t^3)$, which is the slope 2.97 measured in the braking test, but not its constant.

6.3 Executed-only versus per-rollout retraction

This is the June Proposition, restated for both orders and for rollouts that contain the inequality projection. Scheme A integrates the rollouts without retraction, as both manuscripts do; scheme B retracts after every step (phase-space retraction at order 2).

Proposition 5 (Off-manifold rollouts). *Assume (A1) the retraction displacement is bounded by C_R times the residual of c (and of $J_c \dot{q}$ at order 2); (A2) the applied inputs of every sample lie in a bounded set independent of Δt (for instance through two-sided rows), and one step from $\mathcal{M}$ (resp. $T\mathcal{M}$) with such inputs leaves a residual at most $C_c \Delta t^2$; (A3) the rollout step is Lipschitz in the state with constant $1 + L_\Phi \Delta t$, which holds if Π is Lipschitz in q , e.g. under a uniform LICQ for the active rows [8, 9]; (A4) the cost is $S = \sum_t \ell(x_t, \lambda_t) \Delta t + \varphi(x_H)$ with ℓ, φ Lipschitz, where the reaction λ_t is itself Lipschitz in the state for bounded inputs. Then for the same noise realisation:*

(i) $\|x_t^A - x_t^B\| \leq C_R C_c t \Delta t^2 e^{L_\Phi t \Delta t};$

(ii) $|S^A - S^B| \leq \delta := C_R C_c e^{L_\Phi H \Delta t} \Delta t^2 (L_\ell \Delta t H(H+1)/2 + L_\varphi H);$

(iii) the tilted measures satisfy $e^{-2\delta/\beta} \leq dQ^A/dQ^B \leq e^{2\delta/\beta}$, hence $\text{KL} \leq 2\delta/\beta$ and $\text{TV} \leq \frac{1}{2}(e^{2\delta/\beta} - 1)$;

(iv) the projected-average updates satisfy $\|U^{*A} - U^{*B}\| \leq (e^{2\delta/\beta} - 1) \max_k \|\tilde{U}_k^{+B}\| + L_\Pi \max_t \|x_t^A - x_t^B\|$, and the raw-average updates of Algorithm 1 satisfy $\|\hat{v}^A - \hat{v}^B\| \leq (e^{2\delta/\beta} - 1) \max_k \|V_k\|$, a bound that the final projections of their motion parts inherit in the M -norm, while the average of the clamped reactions obeys the first bound with Π the clamped reaction.

At fixed horizon length H every gap is $O(\Delta t^2)$; at fixed horizon time $T_h = H\Delta t$ every gap is $O(\Delta t)$.

Proof. (i) $x_{t+1}^A - x_{t+1}^B = [\Phi(x_t^A) - \Phi(x_t^B)] + [\Phi(x_t^B) - R(\Phi(x_t^B))]$, where the first term is bounded by $(1 + L_\Phi \Delta t) \|x_t^A - x_t^B\|$ and the second by $C_R C_c \Delta t^2$ since x_t^B lies on the manifold; apply the discrete Grönwall inequality. (ii) Sum (i) against the Lipschitz constants. (iii) $dQ^A/dQ^B = e^{-(S^A - S^B)/\beta} Z^B/Z^A$ and each factor lies in $[e^{-\delta/\beta}, e^{\delta/\beta}]$. The TV bound follows from $\text{TV} = \frac{1}{2} \mathbb{E}_{Q^B} |dQ^A/dQ^B - 1|$ and the ratio bound; it is linear in δ , whereas Pinsker's inequality would give only $\sqrt{\delta/\beta}$. (iv) Split $\sum_k w_k^A \tilde{U}_k^{+A} - \sum_k w_k^B \tilde{U}_k^{+B}$ into a weight term, bounded by (iii) sample-wise, and an input term, bounded by the Lipschitz constant of Π and (i). Both schemes average the same raw inputs V_k , so for the raw average the input term vanishes; both project at the same measured state, and the projection is W -nonexpansive. $\square$

Remark 10 (Fixed H versus fixed T_h). The June statement $\delta = C_1 H^2 \Delta t^2$ with an unweighted running cost is $O(T_h^2)$ at fixed horizon time and does not vanish as $\Delta t \rightarrow 0$. With the Δt -weighted cost of (A4) the gap is $O(\Delta t)$ at fixed T_h , which is the statement to use in a rebuttal. The June toy slopes 2.02/2.09/2.10 are consistent with the fixed- H regime; **v4** reproduces them (2.02/2.00/2.23) and finds slopes 0.96/1.00/0.89 (order 1) and 0.92/0.95/0.95 (order 2) at fixed T_h .

Remark 11 (The bounded-input assumption matters). (A2) holds when the applied inputs are bounded, for instance through two-sided rows, and the per-step exploration covariance is fixed. It does not hold for unprojected Gaussian inputs, whose realisations are unbounded; **v4** uses bounded smooth noise for this reason. If exploration is scaled as white noise, $\Sigma \propto \sigma_c^2/\Delta t$, the executed-only rollouts accumulate a mean-curvature drift $\mathbb{E} c_H \rightarrow \frac{1}{2} \sigma_c^2 T_h \text{tr}(N \nabla^2 c N)$ that does not vanish; this is the Itô correction of projected Brownian motion. On the unit sphere **v4** measures $\mathbb{E} \|q_H\|^2 - 1 = 0.0365$ against the predicted $2\sigma_c^2 T_h = 0.036$. Per-rollout retraction, or a Stratonovich-consistent integrator, is then required.

Remark 12 (Cost and tolerance). Per-rollout Gauss–Newton retraction multiplies the rollout cost by $1 + N_{\text{GN}}$ (about three in the June measurements) and buys only the gaps above. A retraction tolerance $\varepsilon > 0$ adds $O(H\varepsilon)$ to (i). Adding the inequality projection to the rollouts costs only assumption (A3); the cost Lipschitz assumption becomes easier because Full PR-MPPI handles obstacles and joint limits without indicator penalties.

7 Averaging

Proposition 6 (Averaging: interpretation and clipping distortion).

- (a) Let $Q^* \propto e^{-S/\beta} P$ over raw sequences V and let Π map V to the applied inputs of its rollout. The raw average $\mathbb{E}_{Q^*}[V]$ minimises the cross-entropy $\text{CE}(Q^*, \mathcal{N}(U, \Sigma))$ over U . The projected average $\mathbb{E}_{Q^*}[\Pi(V)]$ minimises $\text{CE}(\Pi_\# Q^*, \mathcal{N}(U, \Sigma))$, the same projection taken in the space of applied inputs, and the self-normalised estimators $\sum_k w_k V_k$ and $\sum_k w_k \Pi(V_k)$ are consistent for them (Lemma 2).
- (b) At $t = 0$ let Π be the W -projection onto the polyhedron $\mathcal{A}_0$ (so Ω is polyhedral here, not a general convex set), let $\hat{v} = \sum_k w_k v_k$, let $\bar{F}$ be the face of $\mathcal{A}_0$ whose relative interior contains $\Pi(\hat{v})$, call the rows tight at $\Pi(v)$ the active set of v , and let $\mathcal{D}$ be the samples whose active set differs from that of $\hat{v}$. With $\text{Var}_w(v) = \sum_k w_k \|v_k - \hat{v}\|_W^2$, exactly,

$$\sum_k w_k \Pi(v_k) - \Pi(\hat{v}) = \sum_{k \in \mathcal{D}} w_k (\Pi(v_k) - \Pi_{\text{aff } \bar{F}}(v_k)), \quad (9)$$

and consequently

$$\left\| \sum_k w_k \Pi(v_k) - \Pi(\hat{v}) \right\|_W \leq 2 \sum_{k \in \mathcal{D}} w_k \|v_k - \hat{v}\|_W \leq 2 \sqrt{w(\mathcal{D}) \text{Var}_w(v)}. \quad (10)$$

The two averaging schemes coincide when every sample shares the face of $\hat{v}$, including the case where all samples are clipped by the same constraints, and the distortion vanishes as $\beta \rightarrow 0$ when the cost has a unique minimiser among the samples.

Proof. (a) For fixed Σ the Gaussian cross-entropy is minimised by the mean, and the cross-entropy stays finite for the singular measure $\Pi_{\#} Q^*$ although the KL divergence does not. (b) On the region of $\bar{F}$, Π coincides with the affine projection $\Pi_{\text{aff } \bar{F}}$, whose linear part $P_{\bar{F}}$ is W -orthogonal. Hence $\sum_k w_k \Pi_{\text{aff } \bar{F}}(v_k) = \Pi_{\text{aff } \bar{F}}(\hat{v}) = \Pi(\hat{v})$, and subtracting gives (9) because the terms with $k \notin \mathcal{D}$ vanish. Both Π and $\Pi_{\text{aff } \bar{F}}$ are W -nonexpansive and agree at $\hat{v}$, so each remaining term is at most $2\|v_k - \hat{v}\|_W$; Cauchy–Schwarz gives the last bound. $\square$

Remark 13 (Why Algorithm 1 projects once). By (9) the executed command of the earlier controllers, the projected average, is $\Pi(\hat{v})$ plus the distortion. For every row $L_i y \leq b_i$ that binds on $\bar{F}$, $L_i(\Pi(v_k) - \Pi_{\text{aff } \bar{F}}(v_k)) \leq 0$, because $\Pi(v_k)$ satisfies the row and $\Pi_{\text{aff } \bar{F}}(v_k)$ holds it with equality. The distortion therefore moves the command to the feasible side of exactly the rows that bind at $\Pi(\hat{v})$. On the waiter task this cost 16 % of the tracking error and, without anyone choosing it, absorbed the execution error of the real box (Section 13). Algorithm 1 therefore executes the motion part of $\Pi(\hat{v})$, without the distortion, and moves the margin into the rows, where Proposition 11 states what it covers. The reaction is the exception. Along the contact normal the cost is nearly flat and the exploration noise can exceed the reaction set: in the writing task a joint-space noise of 0.8 rad/s^2 moves the reaction of a sample by about 3 N, against a band 2 N wide. The raw average then passes that noise, reduced only by the effective sample size, to the executed reaction, and its reaction component drifts wherever every sample saturates, while averaging after the clamp truncates the noise at the band. With the raw average for the reaction as well, the measured writing force fell from 3.0 to 2.1 N and stayed in the band only half of the time. Separation (Theorem 1) lets the two parts be averaged differently without interference, since the reaction does not move the motion and the motion part does not move the reaction. Without it the scheme fails: with $W = I$ the projection of the raw average corrects the reaction partly through the tangential motion, and the writing force held the band only 17 % of the time, against 98 % for the same metric with the projected average (Section 9).

Remark 14. Item (b) turns the third reviewer concern into a theorem and a diagnostic. The clipped-heterogeneity mass $w(\mathcal{D})$ is available online at no cost, and the raw-versus-projected ablation becomes a check of the bound rather than the only evidence. In v5 identity (9) holds to 10^{-13} and the bound is never exceeded (largest ratio 0.33, median 0.10).

Lemma 2 (Self-normalised update). *Let $g = e^{-S/\beta}$ with $\mathbb{E}_P[g] > 0$ and $\rho := \mathbb{E}_P[g^2]/\mathbb{E}_P[g]^2 < \infty$. Draw $V_k \sim P$ i.i.d. and let $\hat{U}_K = \sum_k g(V_k) \Pi(V_k) / \sum_k g(V_k)$.*

(a) *If $\mathbb{E}_{Q^*}[\|\Pi(V)\|] < \infty$, then $\hat{U}_K \rightarrow \mathbb{E}_{Q^*}[\Pi(V)]$ almost surely.*

(b) *If every applied input is bounded, $|\Pi(V)_i| \leq \bar{u}_i$ for all V and every coordinate i , then*

$$\mathbb{E} \|\hat{U}_K - \mathbb{E}_{Q^*}[\Pi(V)]\|^2 \leq \frac{4\rho \|\bar{u}\|^2}{K}, \quad |\mathbb{E}[\hat{U}_{K,i}] - \mathbb{E}_{Q^*}[\Pi(V)_i]| \leq \frac{(4\rho - 2) \bar{u}_i}{K}. \quad (11)$$

Proof. (a) Apply the strong law of large numbers to the numerator and the denominator. (b) The mean-square bound is the bounded-test-function bound of Agapiou et al. [14] (Thm. 2.1), applied to

$\phi = \Pi(V)_i/\bar{u}_i$ for each coordinate and summed. For the bias, scale g so that $\mathbb{E}_P g = 1$, which leaves $\hat{U}_K$ unchanged and gives $\mathbb{E}_P g^2 = \rho$. Let $|\phi| \leq 1$, $\mu = \mathbb{E}_P[g\phi]$, $B = \frac{1}{K} \sum_k g(V_k)$, $A = \frac{1}{K} \sum_k g(V_k)\phi(V_k)$ and $\hat{\mu} = A/B$. The identity $1/B = 1 - (B-1) + (B-1)^2/B$ gives

$$\mathbb{E}\hat{\mu} - \mu = -\frac{1}{K} \mathbb{E}_P[g^2(\phi - \mu)] + \mathbb{E}[(\hat{\mu} - \mu)(B-1)^2].$$

The first term is at most $2\rho/K$ in magnitude, because $|\phi - \mu| \leq 2$. The second is at most $2\text{Var } B = 2(\rho - 1)/K$, because $|\hat{\mu} - \mu| \leq 2$. Scaling by $\bar{u}_i$ gives the bias bound. $\square$

Remark 15. Bounded applied inputs are the normal case. Two-sided joint-limit CBF rows confine every joint velocity at order 1. At order 2, two-sided HOCBF rows bound the tangential part of the command, and reaction or torque rows bound its normal part. With unbounded Gaussian exploration the raw average has no bounded-test-function guarantee, because its error depends on tilted moments of V . Algorithm 1 clips the noise at $\pm 3\sigma$, which bounds the raw inputs and gives the raw average the same bound (Theorem 2(vi), v9). The constant ρ is the inverse efficiency of the tilt, with effective sample size about K/ρ , so the bound makes the sample budget explicit. Taking Π to be the equality projection recovers the June statement that the projected sampler is a consistent self-normalised estimator of ρ_M^* , read as the pushforward of Q^* under that projection. In v6, a closed-form Gaussian case with $\rho = 5.25$, the projected estimator’s MSE decays as $1/K$ (slope -1.00), stays at 7% of the bound, and is about 40% below the MSE of the raw estimator of its own target.

Table 3: Reviewer concerns and the tool that answers each.

Concern	Answer
Rollouts leave the manifold	Prop. 5, with the fixed- T_h rate and the bounded-input remark
Projected samples are not Gaussian	Projection inside the dynamics (IT-MPC [7]), Prop. 6(a), Lem. 2
Clipping distorts the update	Prop. 6(b) and the raw-versus-projected ablation
Why the inertia metric	Thm. 1(ii)–(iv)
How to choose the safety margin	Lem. 1 and Prop. 4
Why not retract physical constraints	Prop. 2

8 The unified controller

The results above assemble into one controller. Physical equalities enter the Gauss dynamics and carry reaction rows, virtual equalities become motion rows, and every sample is projected in the inertia metric with the separable solver. The update averages the raw inputs. The executed command takes its motion part from the first raw average, projected once at the measured state onto the motion rows tightened by a bound on the execution error, and its reaction from the average of the clamped sample reactions, two choices that separation (Theorem 1) keeps from interfering. Algorithm 1 states one control cycle at order 2.

Two further statements complete the design. The first fixes the order-1 treatment of physical rows, which Proposition 2(b) showed to be fragile.

Proposition 7 (Anchored retraction). *At order 1, retract the physical rows onto the level set $\{c_p = c_p(q_0)\}$ through the measured configuration, and the virtual rows onto $\{c_v = 0\}$. Under the assumptions of Proposition 2(b), with λ_0 the reaction at the measured state, the next static reaction is*

$$\lambda^+ = (\Lambda_K^{-1} + K_{\text{env}}^{-1})^{-1} (K_{\text{env}}^{-1} \lambda_0 + \delta(q^+) - \delta(q_0)) + O(\|\cdot\|^2). \quad (12)$$

Algorithm 1 Unified projection MPPI, one control cycle at order 2

Require: measured $(q_0, \dot{q}_0)$, nominal $U = (u_0, \dots, u_{H-1})$, covariance Σ , K , H , Δt

Require: physical rows c_p , virtual rows c_v with gains K_P, K_D , motion rows, reaction set Ω , metric $W = M$

Require: execution-error bounds $\varepsilon_i \geq 0$: every row $L_i y \leq b_i$ enters as $L_i y \leq b_i - \varepsilon_i \|L_i\|_*$ ($\varepsilon_i = 0$ for model-level claims)

```
1: for  $k = 1, \dots, K$  in parallel do
2:    $x_0^k \leftarrow (q_0, \dot{q}_0)$  and draw  $\epsilon_{0:H-1}^k \sim \mathcal{N}(0, \Sigma)$ , clipped at  $\pm 3\sigma$  per coordinate
3:   for  $t = 0, \dots, H - 1$  do
4:      $\tilde{a} \leftarrow u_t + \epsilon_t^k$ 
5:      $x \leftarrow M$ -projection of  $F\tilde{a}$  onto the motion rows and the stabilised virtual rows, within  $\mathcal{T}$ ; if
       infeasible, the prioritised relaxation of Proposition 8
6:      $\lambda_t^k \leftarrow \Lambda_c^{-1}$ -projection of  $\lambda(\tilde{a})$  onto  $\Omega$  ▷ a clamp for one contact
7:      $a_t^k \leftarrow \Phi^{-1}(x + f, \lambda_t^k)$  ▷ Theorem 1(i)
8:      $x_{t+1}^k \leftarrow$  semi-implicit step of  $\ddot{q} = Fa_t^k + f$  ▷ no retraction, Prop. 5
9:   end for
10:   $S^k \leftarrow \sum_t \ell(x_t^k, \lambda_t^k) \Delta t + \varphi(x_H^k)$ 
11: end for
12:  $w_k \propto e^{-S^k/\beta}$ ,  $\hat{v}_t \leftarrow u_t + \sum_k w_k \epsilon_t^k$  and  $\bar{\lambda}_t \leftarrow \sum_k w_k \lambda_t^k$  ▷ raw average; average of the clamped
    reactions
13:  $x \leftarrow$  line 5 applied to  $\hat{v}_0$  at  $(q_0, \dot{q}_0)$ , and  $a_0^* \leftarrow \Phi^{-1}(x + f, \bar{\lambda}_0)$  ▷ one final projection of the motion
    part
14: execute  $\tau = M(q) a_0^* + b(q, \dot{q})$ 
15:  $U \leftarrow (\hat{v}_1, \dots, \hat{v}_{H-1}, 0)$ , each entry with its reaction set to  $\bar{\lambda}$  ▷ shift and append zero, as PR-MPPI
    Alg. 1
```

For a rigid environment ($K_{\text{env}}^{-1} = 0$) it is $\Lambda_K(\delta(q^+) - \delta(q_0))$: it vanishes for a constant model offset and is first order in the step for a configuration-dependent one. For a compliant environment and a constant offset it contracts to zero geometrically, by the factor $(\Lambda_K^{-1} + K_{\text{env}}^{-1})^{-1} K_{\text{env}}^{-1}$ per cycle.

Proof. Write $q = q^+ + \Delta$ with $K\Delta = J_c^\top \lambda^+$, so $c_p(q) = c_p(q_0) + \Lambda_K^{-1} \lambda^+$ to first order. The contact gives $c_p(q) = \delta(q) - K_{\text{env}}^{-1} \lambda^+$, and the measured state gives $c_p(q_0) = \delta(q_0) - K_{\text{env}}^{-1} \lambda_0$. Eliminating $c_p(q)$ and $c_p(q_0)$ gives the formula. The contraction factor has spectral radius below one because Λ_K and K_{env} are positive definite. $\square$

Remark 16. Anchoring removes the force error but also the force. The anchored order-1 controller can hold a closed chain without internal wrench build-up from calibration error, but without a separate force-feedback law it cannot regulate a contact force. An order-1 controller can regulate force when it adds such a law, for instance $\dot{x} = k_f(F_d - F)$ on a compliant contact; the order-2 controller regulates it through its reaction rows without one.

The second statement makes infeasibility harmless for the reactions.

Proposition 8 (Reaction admissibility is unconditional).

- (a) With $W = M$ and no coupled rows, relaxing the motion rows in any way, for example by slack variables with priorities, leaves the reaction of every sample and of the executed command in Ω .
- (b) With coupled rows, a lexicographic scheme that first enforces the coupled and reaction rows and then minimises the violation of the motion rows keeps every reaction in Ω whenever the coupled and reaction rows are jointly feasible.

Proof. (a) By lines 5–7 of Algorithm 1 and Theorem 1(i), $\lambda(\Phi^{-1}(x + f, \lambda_t^k)) = \lambda_t^k \in \Omega$ for every $x \in \mathcal{T}$, relaxed or not, so the reaction of every sample is admissible whatever the motion rows do. The executed reaction of line 13 is a convex combination of these reactions and lies in Ω because Ω is convex, whatever the motion rows do; for the projected average of Proposition 3(b), $\lambda(\cdot)$ is affine, which gives the same. (b) holds by construction. $\square$

Theorem 2 (Per-cycle guarantees). *Run Algorithm 1 with $\text{rank}[J_p; J_v] = n_p + n_v$, a nonempty compact convex Ω and no coupled rows. At every control cycle:*

- (i) *the model reaction of the executed command at the measured state is admissible, $\lambda(a_0^*) \in \Omega$, whether or not the motion rows are feasible;*
- (ii) *if the tightened motion rows are feasible at the measured state, the executed command satisfies them, and each untightened row holds for every realisation whose argument of that row deviates from the model by at most ε_i ($\varepsilon_i = 0$: the model-level statement);*
- (iii) *at a fixed state, a reaction correction of a sample leaves its motion unchanged and a motion correction leaves its reaction unchanged, so reaction corrections leave every rollout trajectory unchanged; the final projection separates in the same way, but the raw average it acts on depends on the weights, which a correction changes through the rollout cost;*
- (iv) *at rollout steps where no motion row is relaxed, and under the hypotheses of Lemma 1, every HOCBF barrier of the rollout model stays above $-\underline{h}$, so $h_{\text{safe}} \geq \underline{h}$ keeps it nonnegative; Proposition 4 gives the steady-sliding layer and, for zero-order hold on an exact model, the quasi-steady leading term $\Delta t \max(-\ddot{h})/(2k_1k_2)$, which are estimates rather than bounds;*
- (v) *along the rollout model (k the rollout step), at steps where the virtual rows are not relaxed, the virtual residual obeys $(e, \dot{e})_{k+1} = A_\Delta(e, \dot{e})_k + r_k$ with*

$$A_\Delta = \begin{pmatrix} 1 - K_P \Delta t^2 & \Delta t(1 - K_D \Delta t) \\ -K_P \Delta t & 1 - K_D \Delta t \end{pmatrix},$$

which is Schur stable for $0 < K_D \Delta t < 2$ and $0 < K_P \Delta t^2 < 4 - 2K_D \Delta t$. With the deadbeat gains $K_P = 1/\Delta t^2$, $K_D = 1/\Delta t$, exactly

$$e_{k+1} = \Delta t^2 \left(\frac{1}{2} \nabla^2 c_v(\xi_k) [\dot{q}_{k+1}, \dot{q}_{k+1}] - \nabla^2 c_v(q_k) [\dot{q}_k, \dot{q}_k] \right),$$

so $|e_{k+1}| \leq \frac{3}{2} \Delta t^2 \|\nabla^2 c_v\| \bar{v}^2$ from the first step, with $\bar{v}$ bounding the joint velocity including any correction velocity. Under zero-order-hold execution A_Δ becomes $\begin{pmatrix} 1 - \frac{1}{2} K_P \Delta t^2 & \Delta t(1 - \frac{1}{2} K_D \Delta t) \\ -K_P \Delta t & 1 - K_D \Delta t \end{pmatrix}$, which is deadbeat for $K_P \Delta t^2 = 1$, $K_D \Delta t = \frac{3}{2}$;

- (vi) *with the noise clipped at $\pm 3\sigma_i$, the raw inputs are bounded, and with $\bar{u}$ bounding them and $\bar{\ell}$ bounding Ω , the raw average and the average of the clamped reactions estimate $\mathbb{E}_{Q^*}[V]$ and $\mathbb{E}_{Q^*}[\lambda]$ with mean-square errors at most $4\rho \|\bar{u}\|^2/K$ and $4\rho \|\bar{\ell}\|^2/K$; the executed motion part is within $\lambda_{\max}(M)^{1/2}$ times the first root-mean-square error of the projection of its target in the M -norm, and the executed command differs from the scheme with per-rollout retraction by $O(\Delta t)$ at fixed horizon time T_h , with a constant that grows like $e^{L_\Phi T_h} T_h (1 + T_h)$ (Proposition 5).*

Proof. (i) is Proposition 8(a). (ii) is Proposition 3(a). (iii) is Theorem 1(iii) applied to each sample at its state and to the final projection at the measured state; the next state depends only on the motion, so the rollout trajectory is unchanged. (iv) is Lemma 1 at every rollout step whose HOCBF rows are enforced; the estimates are Proposition 4(b),(c). (v) The stabilised virtual row fixes $\ddot{e}_k =$

$-K_D \dot{e}_k - K_P e_k$ at every sampling instant, where $\ddot{e}_k = J_v \ddot{q}_k + \nabla^2 c_v(q_k)[\dot{q}_k, \dot{q}_k]$. The semi-implicit step and Taylor’s theorem give $e_{k+1} = e_k + \Delta t \dot{e}_k + \Delta t^2 \ddot{e}_k + \Delta t^2 (\frac{1}{2} \nabla^2 c_v(\xi_k)[\dot{q}_{k+1}, \dot{q}_{k+1}] - \nabla^2 c_v(q_k)[\dot{q}_k, \dot{q}_k])$ and $\dot{e}_{k+1} = \dot{e}_k + \Delta t \ddot{e}_k + O(\Delta t^2)$, which is the recursion. The Jury conditions of its characteristic polynomial $z^2 - (2 - K_P \Delta t^2 - K_D \Delta t)z + (1 - K_D \Delta t)$ give the stability region. With the deadbeat gains the first three terms of e_{k+1} cancel, which leaves the exact curvature term. Under zero-order hold the step is $e_{k+1} = e_k + \Delta t \dot{e}_k + \frac{1}{2} \Delta t^2 \ddot{e}_k + O(\Delta t^3)$, and the trace and determinant of the resulting matrix vanish exactly at the stated gains. (vi) Each entry of U entered as zero and received at most $H - 1$ updates, each a convex combination of clipped noise and so at most $3\sigma_i$ per coordinate, and the reset of line 15 keeps its reaction component in the bounded set Ω ; its sample adds one more clipped perturbation, so the raw inputs are bounded. Lemma 2(b) with Π the identity then bounds the raw average, whose target is the cross-entropy minimiser of Proposition 6(a), and with Π the clamped reaction it bounds $\bar{\lambda}$. The motion part is the M -projection of $F \hat{v}_0$, and F and the projection are both M -nonexpansive, so its distance to the projection of $F \mathbb{E}_{Q^*} V_0$ is at most $\lambda_{\max}(M)^{1/2} \|\hat{v}_0 - \mathbb{E}_{Q^*} V_0\|$. The last statement is Proposition 5(iv) in its raw-average form, with $H = T_h/\Delta t$. $\square$

Remark 17 (Separation does not survive the weights). Item (iii) cannot be extended to the executed command with the weights recomputed. Two samples with motions 0 and 1 and costs (0, 4) give the weighted motion $e^{-4}/(1 + e^{-4}) = 0.018$ at $\beta = 1$. If a reaction correction lowers the second cost to 1 without touching either motion, the weighted motion becomes $e^{-1}/(1 + e^{-1}) = 0.269$. Separation is a property of each projection and of averages with given weights; the admissibility of the executed command in (i) and (ii) does not depend on the weights and holds regardless.

Remark 18 (What the guarantees are about). Items (i)–(iii) and (v) concern the controller’s model at the sampling instants. The measured reaction and motion differ from it by the model error, the tracking error of the low-level loop and the zero-order hold. In the writing task, for instance, the commanded reaction stayed in the band throughout, and the measured one 99.3% of the time. Proposition 11 in Section 11 turns a bound on this difference into a margin of the rows, which Algorithm 1 applies with a calibrated ε on the waiter task.

Remark 19 (The order-1 member and coupled rows). At order 1 the same cycle uses the nested projection $\Pi^{(1)}$, Euler rollouts, and at execution the Gauss–Newton retraction of the virtual rows with the anchored retraction of the physical rows (Proposition 7). With coupled rows (torque limits, passive bodies) the two projections of lines 5–7 merge into one joint projection, and Proposition 8(b) replaces (i).

9 Simulation: curved-surface writing

The controllers were re-implemented from the two manuscripts, since the original code is not available. They run on the curved-surface writing task of CAMP-MPPI (Sec. IV-B), with the offset of the board model as the swept variable.

Setup. The arm is a Franka-like 7-DoF model with the datasheet kinematics and joint limits, the inertial parameters of Gaz et al., and 0.1 kg m² armature. The controller model is a batched implementation that reproduces MuJoCo’s mass matrix, bias forces and Jacobians to 2×10^{-15} . The physics is MuJoCo at 1 kHz with an explicit contact law against the real board, of stiffness 2×10^4 N/m, damping 60 Ns/m and Coulomb friction $\mu = 0.2$. The joints carry a viscous friction of 1 N m s/rad that no controller models. The board is a sphere of radius 0.6 m, the pen traces a 16×8 cm lemniscate in 8 s, and the admissible contact band is 2–4 N. All controllers run at 100 Hz with $H = 15$ and $K = 256$, as in CAMP-MPPI. PR-MPPI commands joint velocities ($\sigma = 0.1$ rad/s), tracked by a computed-torque

loop of 10 Hz bandwidth. The order-2 controllers command accelerations ($\sigma = 0.8 \text{ rad/s}^2$) under HOCBF joint limits ($k_1 = k_2 = 4$) and velocity limits. The board model is offset from the real board by $\delta \in \{-1, -0.5, 0, 0.5, 1\}$ mm, negative δ placing it below the real surface, and every condition runs with three seeds.

Implementation notes. Two details decide whether the order-2 controllers work at all. With the pure feed-forward command $\tau = Ma + b$, sampling noise in the null space of the task, such as joint 7 about the pen axis, random-walks until the arm leaves the path; joint-velocity limits as motion rows and a joint-velocity cost ($w_v = 5$) remove it. The low-level loop of PR-MPPI interpolates its reference over the control period, so its effective stiffness in Proposition 2(b) is the period average $M(\omega^2/2 + 2\omega/\Delta t)$ rather than $M\omega^2$.

Controllers. PR-MPPI retracts all equalities, as published. PR-MPPI anchored applies Proposition 7. CAMP-MPPI keeps the pen orientation as a cost; CAMP-MPPI manifold places it inside the Gauss manifold, as if the board supplied the orienting torque. The unified controller is Algorithm 1, with $\varepsilon = 0$: the claims below concern the commanded reaction, and the measured force is reported as it comes. CAMP-MPPI and PR-MPPI run as published, with the average of the projected samples. Unified-joint replaces its separable solver by the joint projection, and Unified $W = I$ replaces its metric.

Table 4: Mean contact force [N] against the offset of the board model (three seeds).

Controller	$\delta = -1$ mm	-0.5 mm	0	$+0.5$ mm	$+1$ mm
Prop. 2(b), PR-MPPI	15.94	7.97	0 (lift-off)	0 (lift-off)	0 (lift-off)
PR-MPPI	15.97	7.98	0.00	0.00	0.00
PR-MPPI anchored	0.00	0.00	0.00	0.00	0.00
CAMP-MPPI (cost)	3.04	3.04	3.04	3.04	3.04
Unified	3.03	3.03	3.03	3.03	3.03
Unified $W = I$	3.28	3.28	3.28	3.28	3.28

Table 5: Performance at $\delta = 0$ over three seeds of 8 s: completed runs, with the mean divergence time of the others in parentheses. Metrics cover the executed portion; order-2 results are identical for every δ .

Controller	In band [%]	Contact [%]	Tracking [mm]	Pen tilt [deg]	Completed
PR-MPPI	0.0	0.0	1.09	0.00	3/3
PR-MPPI anchored	0.0	0.0	1.09	0.00	3/3
CAMP-MPPI (cost)	99.1	100.0	1.77	0.72	3/3
CAMP-MPPI (manifold)	42.1	100.0	20.99	25.54	0/3 (4.1 s)
Unified	99.3	100.0	1.58	0.11	3/3
Unified-joint	99.3	100.0	1.58	0.11	3/3
Unified $W = I$	16.8	53.1	1.87	0.18	3/3

Findings.

1. Proposition 2(b) predicts the force of PR-MPPI. It pressed with 15.97 N at $\delta = -1$ mm and 7.98 N at -0.5 mm, against the predicted 15.94 and 7.97 N, and it lost contact for $\delta \geq 0$. The prediction uses the period-averaged stiffness $K_{\text{eff}} = M(\omega^2/2 + 2\omega/\Delta t)$ of the interpolated computed-torque loop; the proportional stiffness $M\omega^2$ alone underestimates the slope by a factor of 1.5.

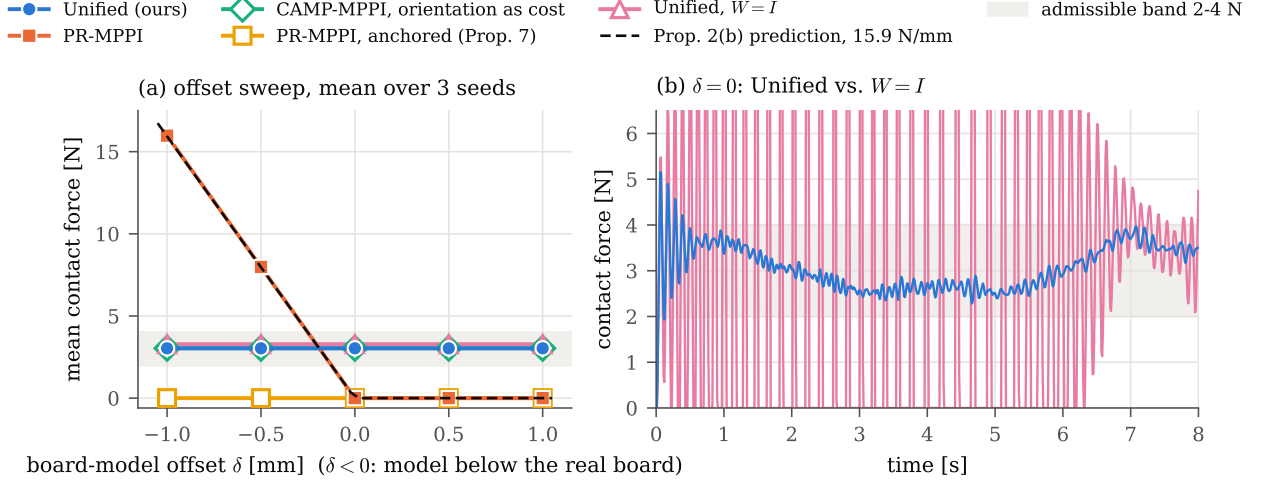


Figure 1: (a) Mean contact force against the board-model offset; the dashed line is Proposition 2(b) with the effective stiffness of the low-level loop, and the shaded band is the admissible range. (b) Contact force at $\delta = 0$ for the unified controller and its $W = I$ ablation.

2. Anchoring (Proposition 7) removed the offset-driven force for every δ , and with it every contact: the anchored order-1 controller has no force feedback and cannot write.
3. The order-2 controllers are offset invariant, as Proposition 2(a) states: their mean force varies by at most 8.9×10^{-16} N across the five offsets.
4. The separable and joint solvers produced identical force traces (largest difference 2.4×10^{-11} N), as Theorem 1(iii) requires. Replacing the inertia metric by $W = I$ lowered the in-band fraction from 99.3% to 16.8% (lower in every seed), with a force standard deviation of 4.4 N against 0.45 N. Without separation, the projection of the raw average corrects the reaction partly through the tangential motion, so the exploration noise along the contact normal (remark after Proposition 6) reaches the pen; with the average of the projected samples the same change of metric cost only 1.6 points (99.3 to 97.8%). The averaging of Algorithm 1 relies on the inertia metric.
5. Treating the virtual orientation as physical failed as Proposition 2 predicts: CAMP-MPPI with the orientation in its manifold diverged in all 3 runs, at 4.1 s on average, and until then it held the band only 42% of the time with a tilt of 26 deg. Keeping the orientation as a cost worked once the null space was damped, with a tilt of 0.7 deg (RMS over seeds) against 0.11 deg for the virtual rows of the unified controller. Its tilt depends on the weight: for seed 0 it was 0.71, 0.09 and 0.04 deg at $w_o = 5 \times 10^3$, 5×10^4 and 5×10^5 , with tracking errors of 1.8, 1.8 and 2.6 mm. The virtual rows need no weight.
6. Every executed reaction of the unified controller stayed in the band (commanded range 2.12–3.39 N), as Theorem 2(i) states; the priority fallback of Proposition 8 was needed in 0 of its projections.

10 Simulation: dual-arm closed chain

This section re-implements the dual-arm experiment of PR-MPPI (Sec. III-B, Eq. (17)) and adds a calibration error.

Setup. Two Franka-like arms stand 0.7 m apart and hold a 1 kg tray at its ends with both tools pointing down. The grasps are explicit 6-DoF springs (3×10^4 N/m, 300 N m/rad), so the internal wrench $W_{\text{int}} = (W_l - W_r)/2$ is measured exactly; MuJoCo runs at 2 kHz and the joints carry viscous friction. The equality has six physical rows (the relative pose of the tools, enforced by the tray) and two virtual rows (tray roll and pitch). The tray is lowered from (0.60, 0, 0.40) to (0.42, 0, 0.20) m by joint tracking of an IK solution, which drives joint 4 of both arms from -1.70 to -2.52 rad. As in PR-MPPI, an artificial lower bound is placed 0.06 rad above the goal value, with a 0.05 rad CBF margin, and the ablation without inequality projection carries a joint-limit penalty one tenth of the tracking weight. All controllers run at 50 Hz with $H = 15$ and $K = 256$; PR-MPPI commands joint velocities through a 10 Hz computed-torque loop, and the unified controller commands accelerations with a reaction box of 5 N and 1 N m on the internal wrench. In E2 the controller’s model of the right-arm base is displaced by δ_b along y , while the grasp transform keeps its nominal value.

Table 6: E1: joint-limit stress test without calibration error (three seeds). Residuals are measured on the physical arms; the command residual is the largest entry of the model equality at the retracted command.

Controller	Bound viol. [rad]	Margin pen. [rad]	Command residual	Chain [mm / mrad]	Tilt [deg]	Internal force [N]
PR-MPPI	0.000	0.000	4.1×10^{-14}	0.05 / 0.23	0.03	0.47
PR-MPPI w/o retraction	0.000	0.000	0.111	0.20 / 1.57	3.61	0.53
PR-MPPI w/o inequality	0.059	0.109	4.1×10^{-14}	0.05 / 0.25	0.03	0.50
Unified	0.000	0.000	–	0.67 / 3.89	1.76	2.00
Unified, $\varepsilon = 0$	0.008	0.058	–	0.54 / 2.93	1.70	1.08
Shield-MPPI	0.000	0.003	–	0.35 / 1.70	1.70	1.41
DualGuard MPPI	0.000	0.001	–	0.50 / 2.76	1.69	1.52
GS-MPPI	0.002	0.052	–	0.35 / 1.90	1.70	1.09

Table 7: E2: mean internal force [N] against the calibration error δ_b of the right-arm base (three seeds, $t \geq 1$ s).

Controller	$\delta_b = 0$	0.5 mm	1 mm	2 mm
Prop. 2(b), closed chain	0.46	4.22	7.98	15.50
PR-MPPI	0.47	4.15	7.84	15.23
PR-MPPI anchored	0.47	0.45	0.45	0.45
Unified	2.00	2.00	2.00	2.00

Findings.

1. E1 reproduces the pattern of PR-MPPI Table II. Full PR-MPPI never crossed the bound and returned commands that satisfy the model equality to 4.1×10^{-14} . Without retraction the command residual grew to 0.11 and the tray tilted by 3.6 deg on average; without inequality projection the cost penalty let joint 4 cross the bound by 0.059 rad (the manuscript reports 0.022 rad). The unified controller, with its joint rows tightened by the calibrated execution-error bounds below, kept the bound, using 0.000 of the 0.05 rad margin, and held the internal wrench inside its box (2.00 N); its tray tilt of 1.8 deg is larger than that of PR-MPPI (0.03 deg) because the 1 kg tray is unmodelled and the command is pure feed-forward.
2. The internal force of PR-MPPI grew linearly with the calibration error, as the closed-chain form of Proposition 2(b) predicts: the zero-error baseline plus $(C_l + C_r + 2C_g)^{-1}\delta_b$, with the

compliances of the two interpolated computed-torque loops and of the grasp springs at the final configuration (7.52 N/mm), matches the measurement to within 2%. Anchoring (Proposition 7) removed it: 0.45 N at every δ_b . The unified controller is exactly invariant (2.00 N at every δ_b), since a translation of the base changes no Jacobian and the Gauss dynamics use nothing else (Proposition 2(a)).

3. The margin of Algorithm 1 is needed here as well. Without it ($\varepsilon = 0$) the unified controller crossed the bound by 0.008 rad in every run: the command rides the barrier, and the unmodelled tray makes the realized acceleration of joint 4 lag the command by about 0.9 rad/s² in steady state. The bounds are calibrated like those of the waiter, as 1.5 times the largest period-averaged deviation of each joint acceleration from the model in separate runs of the untightened controller ($\delta_b \in \{0, 2\}$ mm, seeds 100 and 101): 2.3–3.8 rad/s² for the shoulder and elbow joints and up to 18 rad/s² at the wrists, where the grasp springs oscillate. They are conservative: the tightened controller stopped 0.26 rad from the goal, against 0.115 rad for PR-MPPI and 0.066 rad without the margin; modelling the tray load would shrink them (open item).
4. The warm start must append a zero input, as in Algorithm 1 of PR-MPPI. Repeating the last input instead carried the tracking velocity past the goal and drove joint 4 0.6 rad beyond the bound in the ablation without inequality projection, which the CBF rows of the full controller had masked.

11 Certified wrench constraints under model uncertainty

Hard wrench constraints are only as good as the interaction model behind them: the mass, centre of mass and friction of a passive object are rarely known. This section makes the reaction and support rows robust to a parameter set, keeps the certificate of Theorem 2, and lets the set shrink online.

Setting. At a rollout state, an interaction wrench is affine in the command for fixed parameters, $\lambda(a; \theta) = G(\theta)a + g(\theta)$, with admissible set $\mathcal{K}(\theta) = \{\lambda : A(\theta)\lambda \leq b(\theta)\}$ and θ in a parameter set Θ . The robust admissible set is

$$\mathcal{A}_\Theta = \{a : A(\theta)(G(\theta)a + g(\theta)) \leq b(\theta) \quad \forall \theta \in \Theta\}. \quad (13)$$

Proposition 9 (Vertex reduction). *Let $\Theta \subset \mathbb{R}^p$ be a box, and let every row $r_i(a, \theta) = A_i(\theta)(G(\theta)a + g(\theta)) - b_i(\theta)$ be convex in each coordinate θ_j separately for every fixed a ; rows multilinear in θ (affine in each coordinate) are the main case. Then $\mathcal{A}_\Theta = \bigcap_{v \in \text{vert } \Theta} \mathcal{A}_{\{v\}}$, a polyhedron with at most 2^p copies of each row, and the projection machinery of Sections 4–8 applies unchanged. If the rows are jointly affine in θ , the same holds for any polytope Θ with its vertices. For the rows tightened as in Proposition 11 below, the same holds if in addition $A_i(\theta)$ is affine in each coordinate and ε is one fixed bound for all $\theta \in \Theta$, because $\theta_j \mapsto \|A_i(\theta)\|_*$ is then the norm of an affine function; alternatively, assume the tightened residual itself is convex in each coordinate.*

Proof. For fixed a , a function convex in one coordinate attains its maximum over an interval of that coordinate at an endpoint. Applying this coordinate by coordinate moves any maximiser over the box to a vertex without lowering the value, so $r_i(a, \theta) \leq 0$ for all $\theta \in \Theta$ is equivalent to $r_i(a, v) \leq 0$ at every vertex. $\square$

Proposition 10 (Certificate). *Let the true parameter lie in Θ . Keep the rows of $\mathcal{A}_\Theta$ hard at the measured state, relaxing the other motion rows first if needed (Proposition 8(b)), and let them be feasible there. Let a_0 be the command executed by Algorithm 1. Then the wrench that the model with the true parameter predicts for a_0 is admissible: $\lambda(a_0; \theta_{\text{true}}) \in \mathcal{K}(\theta_{\text{true}})$.*

Proof. Every sample's first input is projected into $\mathcal{A}_\Theta$, which is convex and contained in $\mathcal{A}_{\{\theta_{\text{true}}\}}$, so Proposition 3 places the weighted average in it. $\square$

In the waiter task below the rows of $\mathcal{A}_\Theta$ are motion rows, since the task has no physical equality, so Theorem 2(i) does not apply to them. They are feasible whenever the tray can be stopped, because zero tray acceleration with a level tray satisfies every support row.

The certificate concerns the model wrench at the sampling instant. The wrench that the object actually receives differs from it by the execution error of the low-level loop, by the change of the state under the zero-order hold, and by unmodelled dynamics. A margin in the rows absorbs this difference.

Proposition 11 (From the model wrench to the realized wrench). *Write the realized wrench over a control period as $\lambda_{\text{real}}(\tau) = \lambda(a_0; \theta_{\text{true}}) + \delta(\tau)$, and tighten the rows of $\mathcal{A}_\Theta$ to $A_i(\theta)\lambda \leq b_i(\theta) - \varepsilon \|A_i(\theta)\|_*$, with $\|\cdot\|_*$ the dual norm.*

- (a) *If $\|\delta(\tau)\| \leq \varepsilon$ at every τ of the period, then $\lambda_{\text{real}}(\tau) \in \mathcal{K}(\theta_{\text{true}})$ over the whole period.*
- (b) *If only the period average satisfies $\|\bar{\delta}\| \leq \varepsilon$, then the period-averaged realized wrench $\bar{\lambda}_{\text{real}}$ lies in $\mathcal{K}(\theta_{\text{true}})$.*

Proof. By Proposition 10 applied to the tightened rows, $A_i \lambda(a_0; \theta_{\text{true}}) \leq b_i - \varepsilon \|A_i\|_*$. By Hölder's inequality $A_i \delta \leq \|A_i\|_* \|\delta\| \leq \varepsilon \|A_i\|_*$, pointwise for (a) and for $\bar{\delta}$ for (b), because $\bar{\lambda}_{\text{real}} = \lambda(a_0; \theta_{\text{true}}) + \bar{\delta}$. $\square$

Part (b) is weaker than (a): an average inside the margin does not bound the instantaneous error. For a rigid object on a support it is still the relevant quantity for tipping over a short period, because the object's rotation responds to the integrated torque, but a claim about every instant needs (a).

Proposition 12 (Set-membership adaptation). *Let the measurements be linear in a parametrisation ϕ of the model, $y_k = \Psi_k \phi + e_k$, and let $\theta = \eta(\phi)$ with η defined on a set $\hat{P}_0$. Let $\phi_{\text{true}} \in \hat{P}_0$ and $\theta_{\text{true}} \in \Theta_0$. Assume $|e_k| \leq \bar{e}_k$ componentwise whenever $\phi_{\text{true}} \in \hat{P}_{k-1}$ and the object sticks during window k . The bound $\bar{e}_k$ may depend on $\hat{P}_{k-1}$, and it must cover errors of the regressor Ψ_k through a supremum over $\hat{P}_{k-1}$. In sticking windows let $\hat{P}_k$ be any set containing $\hat{P}_{k-1} \cap \{\phi : |y_k - \Psi_k \phi| \leq \bar{e}_k\}$, for instance the exact intersection or a box enclosing it, and otherwise let $\hat{P}_k = \hat{P}_{k-1}$. Let*

$$\Theta_k = \Theta_{k-1} \cap \square \eta(\hat{P}_k), \quad (14)$$

where $\square$ denotes any box enclosure. Then $\phi_{\text{true}} \in \hat{P}_k$ and $\theta_{\text{true}} \in \Theta_k \subseteq \Theta_{k-1}$ for every k . Proposition 10 therefore holds at every cycle with Θ_k , while the conservatism of $\mathcal{A}_{\Theta_k}$ decreases monotonically.

Proof. By induction on k , from the base case $\phi_{\text{true}} \in \hat{P}_0$, $\theta_{\text{true}} \in \Theta_0$. If $\phi_{\text{true}} \in \hat{P}_{k-1}$ and window k is used, the bound $\bar{e}_k$ holds for the true error, so ϕ_{true} satisfies the k -th measurement constraint and lies in $\hat{P}_k$. Then $\theta_{\text{true}} = \eta(\phi_{\text{true}}) \in \square \eta(\hat{P}_k)$, and $\theta_{\text{true}} \in \Theta_{k-1}$ by the induction hypothesis. The intersection with Θ_{k-1} makes the boxes nonincreasing by construction. $\square$

The intersection with Θ_{k-1} is not a formality. A box enclosing a shrinking set need not shrink when the enclosure is itself an outer approximation, as with interval arithmetic. It does shrink when $\square$ is the exact interval hull of $\eta(\hat{P}_k)$ and $\hat{P}_k$ is nonincreasing, which is the case below.

Remark 20 (When to measure). The linear measurement model holds only while the object sticks to the support. The robust rows certify only the model wrench at the sampling instants. The object sticks when, in addition, the realized deviation stays within the margin of Proposition 11. Updates are therefore taken only in windows where a measured proxy of sticking holds; below it is the measured centre of pressure, which does not check slipping (the slip rows never bind in the waiter task). The bound $\bar{e}_k$ must cover sensor noise and model error, such as the object's rocking on a compliant contact and the uncertainty in where it stands on the support. An empty $\hat{P}_k$ proves that a bound was violated; the update is then rejected and counted.

Waiter task. A tall, narrow box (base half-width $r = 2$ cm, height 24 cm) stands on a tray carried by the arm. Let n be the tray normal, $u_1, \dots, u_8$ eight directions in the tray plane, $c = \cos(\pi/8)$, and $s = a_B - g$ the acceleration of the tray point under the box relative to gravity. The support rows are $h u_k^\top s \leq r c n^\top s$ (no tipping: the centre of pressure stays in a regular octagon inside the base), $u_k^\top s \leq \mu c n^\top s$ (no slipping) and $n^\top s \geq 3 \text{ m/s}^2$ (no lift-off). They are rows in the joint acceleration because s is affine in it, and the tray orientation (level and heading) enters as three hard virtual equalities. Each row is affine in h or in μ for a fixed command, so Proposition 9 applies with the two end points of each interval, 33 rows in all. The row vectors $h u_k - r c n$ and $u_k - \mu c n$ acting on s are affine in each parameter, which the tightening below needs.

The rows carry two margins. The margin of Proposition 11 is explicit: Algorithm 1 tightens every support row $L_i s \leq b_i$ to $L_i s \leq b_i - \varepsilon \|L_i\|$, where $\varepsilon = 0.165 \text{ m/s}^2$ bounds the period-averaged deviation of the realized s from the model and is calibrated below. Since the wrench on the box is m times s to within the rotation of the tray, this is the tightening of the proposition with εm in wrench units. The row vectors are affine in each parameter and ε is the same for every $\theta \in \Theta$, which is the tightened case of Proposition 9: the tightened rows are convex in each parameter, and the end points of the intervals still suffice. The second margin is geometric and comes on top. The true no-tipping condition is that the centre of pressure stays in the square base, and the octagon lies inside the square with a gap of $(1 - c)r$ along each base axis, which absorbs a further $(1 - c)rg/h = 0.11 \text{ m/s}^2$ of tangential deviation at $h = 0.14 \text{ m}$ along those axes.

A wrist force–torque sensor measures the contact wrench on the box about the tray point, with uniform noise of 0.1 N and 3 mN m per 1 kHz sample. Measurements are taken over sliding windows of $T_w = 100 \text{ ms}$, after a 0.1 s settling time, and only while the measured centre of pressure, averaged over the 20 ms control period, stays within 90% of the base: this is the sticking condition of the remark. Three adaptive variants use three measurement models, all linear in the unknown.

Direct (F/T only). While the box sticks, its impulse over a window is linear in $\phi = (m, mh)$: $Y_F = m\nu$ and $Y_\tau = (mh)n \times \nu + m d \times \nu$, with ν the change of the tray-point velocity relative to gravity over the window and d the unknown position of the box on the tray. The torque bound is $\bar{\varepsilon}_\tau$ plus $\bar{m} \bar{d} |\nu|$, with $\bar{m}$ the current upper bound on m ; it depends on the current set, as Proposition 12 allows. Here $\eta(\phi) = \phi_2/\phi_1$ is defined because the prior keeps $m \geq 0.2 \text{ kg}$. The offset term is needed because under sustained tangential load the box creeps on the tray, by up to 1.7 mm per run in the simulator’s soft contact, and it dominates the bound.

Box tracked. The same model, with the position of the box tracked to 0.2 mm (for instance by a camera): the torque is taken about the tracked position and $\bar{d} = 0.5 \text{ mm}$.

COP differences (F/T only). The centre of pressure over a window is $p_c = (n \times Y_\tau)/(n^\top Y_F) = d - h \nu_t/\nu_n$, so for two windows $j < k$

$$p_c(k) - p_c(j) = -h (\kappa_k - \kappa_j) + (d_k - d_j), \quad \kappa = \nu_t/\nu_n, \quad (15)$$

which is linear in h ($\phi = h$, η the identity), with κ known from the arm kinematics and the unknown position cancelled up to its creep between the windows. The error bound is the sum of the two centre-of-pressure errors, obtained from the force and torque bounds by the quotient bound, and of the creep bound $|d_k - d_j| \leq d_0 + \dot{d}_{\max}(t_k - t_j + T_w)$; windows up to 0.6 s apart are paired. In all three cases the parameter set is an interval, updated exactly, so its hull shrinks monotonically as Proposition 12 requires.

Calibration of the bounds. The bounds are not set by hand. Separate runs of the oracle at $h \in \{0.05, 0.115, 0.18\} \text{ m}$, the ends and the middle of the prior, with seeds not used in the evaluation, give the largest residual of each model under the true parameters and the true box position. Each bound is 1.5 times that value: $\bar{\varepsilon}_F = 0.033 \text{ N}$, $\bar{\varepsilon}_\tau = 3.5 \text{ mN m}$, $\bar{d} = 2.5 \text{ mm}$, $\dot{d}_{\max} = 1.7 \text{ mm/s}$ and $d_0 = 0.1 \text{ mm}$. The calibration runs are separate from the simulation runs used for evaluation. The

torque error is dominated by the small rocking of the box on its compliant contact, which the rigid model neglects. Its impulse is a change of angular momentum that does not grow with the window, so the error per unit time falls as $1/T_w$: the largest calibration residual drops from 5.3 to 2.3 mN m between 40 and 100 ms windows, which is why the windows are 100 ms long. The force bound is smaller than the per-sample noise bound because a window averages 100 independent samples. Like the others, it is therefore a calibrated assumption rather than a worst case, and the evaluation below checks it. The execution-error bound ε is calibrated in the same way, but with the untightened oracle, which moves most aggressively: over the same heights and seeds the period average of $\|s_{\text{real}} - s_{\text{cmd}}\|$ reached at most 0.110 m/s², so $\varepsilon = 0.165$ m/s². The deviation grows with the commanded acceleration, so this bound, set by the lowest centre of mass, is conservative for taller boxes.

The simulation uses MuJoCo at 1 kHz with an elliptic friction cone and real contact between box and tray. The arm executes $\tau = Ma + b + D\dot{q} - \tau_{\text{load}}$, with the known joint damping $D\dot{q}$ and the measured load τ_{load} of the previous 1 ms step, so the executed acceleration is the commanded one up to the error that Proposition 11 covers.

The true object has $h = 0.14$ m, $m = 0.6$ kg and $\mu = 0.6$, so its tipping limit is $(r/h)g = 1.40$ m/s² along the base axes. The reference moves the tray by 0.4 m in 0.8 s, holds for 1.2 s and moves back, with a peak acceleration of 3.6 m/s². The nominal controller assumes $h = 0.07$ m, so its tightened rows admit 2.4 m/s². The robust controller uses $h \in [0.05, 0.18]$ m and $\mu \in [0.4, 0.8]$, so its rows admit only 0.84 m/s². The three adaptive controllers start from the same intervals and shrink the interval of h online. The oracle knows h and μ , and its rows admit 1.13 m/s². All run Algorithm 1 at 50 Hz with $H = 20$ (0.4 s), $K = 256$ and the same cost, on the GPU implementation of Section 12, five seeds each. A sweep repeats the comparison for $h \in \{0.08, 0.11, 0.17\}$ m with three seeds, using the two better adaptive variants.

Table 8: Waiter task with the true $h = 0.14$ m, over 5 seeds: tipped runs, tracking error of each move (safe runs), time to come within 1 cm of each goal after the move starts, peak horizontal tray acceleration, and the interval of h used by the rows at the end (union over seeds).

Controller	Tipped	RMSE [mm]		Reach [s]		Peak $ a_h $ [m/s ²]	Final h [m]
		move 1	move 2	move 1	move 2		
Nominal ($h = 0.07$ m)	5/5	–	–	–	–	2.36	0.07
Robust	0/5	39.2	41.2	0.94	0.95	0.96	0.050–0.180
Adaptive, direct F/T	0/5	38.0	39.3	0.94	0.93	0.99	0.106–0.174
Adaptive, COP differences	0/5	39.1	30.4	1.08	0.88	1.18	0.131–0.149
Adaptive, box tracked	0/5	28.0	29.8	0.88	0.87	1.19	0.131–0.149
Oracle	0/5	25.7	27.1	0.86	0.86	1.28	0.14

Table 9: Sweep over the true centre-of-mass height, all inside the prior interval $[0.05, 0.18]$ m: tipped runs and the tracking error of move 2 [mm] over the safe runs. The last row is the acceleration that the rows of the oracle admit.

Controller	$h = 0.08$ m	$h = 0.11$ m	$h = 0.14$ m	$h = 0.17$ m
Nominal ($h = 0.07$ m)	0/3, 20.5	2/3, 20.4	5/5	3/3
Robust	0/3, 42.0	0/3, 42.0	0/5, 41.2	0/3, 42.0
Adaptive, COP differences	0/3, 21.3	0/3, 23.0	0/5, 30.4	0/3, 42.0
Adaptive, box tracked	0/3, 21.2	0/3, 22.9	0/5, 29.8	0/3, 42.0
Oracle	0/3, 20.9	0/3, 23.1	0/5, 27.1	0/3, 38.1
Oracle rows admit [m/s ²]	2.10	1.48	1.13	0.90

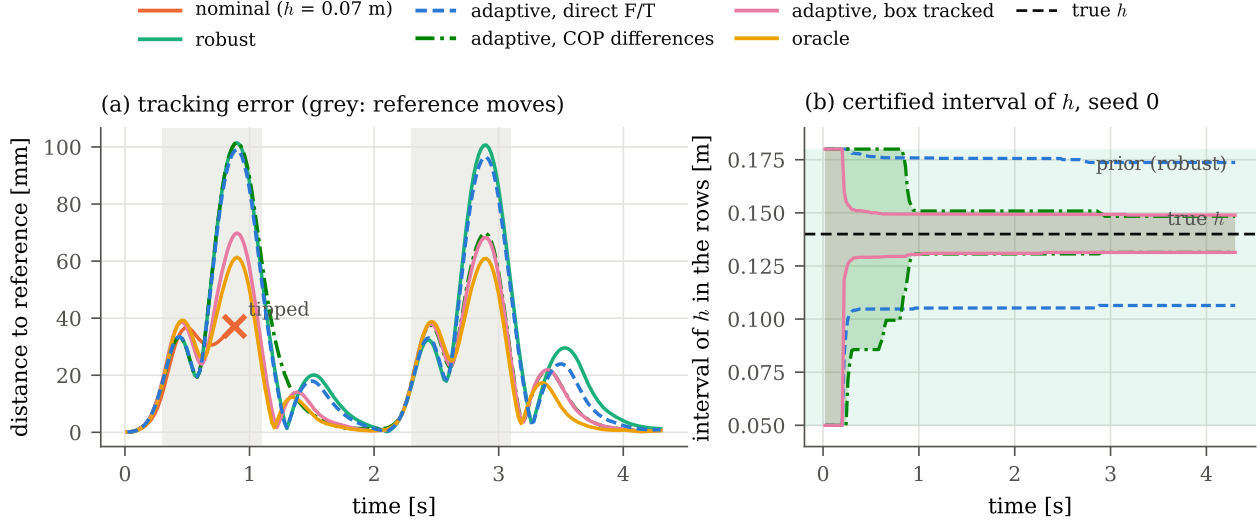


Figure 2: Waiter task at $h = 0.14$ m, seed 0. (a) Distance of the tray point to the reference; a cross marks the moment the nominal controller tipped the box, and the direct adaptive controller (dashed) nearly coincides with the robust one. (b) The interval of the centre-of-mass height used by the rows of the three adaptive controllers, inside the prior interval of the robust controller, against the true value.

Findings.

1. The nominal model tipped the box in 5 of 5 runs at $h = 0.14$ m, 0.56 s into the first move on average, and in 0/3 at $h = 0.08$, 2/3 at $h = 0.11$, 3/3 at $h = 0.17$ of the sweep. Its peak horizontal acceleration of 2.36 m/s^2 is admissible for the assumed $h = 0.07$ m but not for the true box.
2. No controller with certified rows tipped the box: 0 of 61 runs over all heights. This is consistent with Propositions 10 and 11: the certificate covers the model wrench for any h in the prior interval, and the tightening covered the measured deviation (item 3). The price of the robust rows is speed: at $h = 0.14$ m they held the peak acceleration to 0.96 m/s^2 and tracked move 2 with 41.2 mm RMSE, against 1.28 m/s^2 and 27.1 mm for the oracle.
3. What the certified runs verify, level by level (Propositions 10 and 11). At the sampling instants the model wrench under the true h used at most 93% of the octagon: the certificate. Averaged over each 20 ms period, the realized tray acceleration deviated from the model by at most 0.115 m/s^2 over all certified runs, 70% of the calibrated ε , and the period-averaged centre of pressure stayed within 87% of the edge of the base: the hypothesis and the conclusion of Proposition 11(b) hold. At the 1 ms resolution of the simulation the hypothesis of (a) does not: in a diagnostic run of the oracle at $h = 0.14$ m, the deviation exceeded ε in 1.6% of the settled substeps (99th percentile 1.08 and maximum 1.69 times ε), and the instantaneous centre of pressure touched an edge in 28% of the control periods; in the sweep it touched an edge in some substep of 60 of 61 certified runs. The pointwise statement (a) is therefore not verified with this bound. The box rocked momentarily in those substeps but never tipped; the nominal model's wrench under the true h reached 1.37 times the octagon before its box tipped.
4. The interval of h contained the true value in every adaptive run, and no measurement contradicted the bounds (0 rejections), as Proposition 12 states. How far it shrinks is set by the

measurement model. The direct model ended at $[0.106, 0.174]$ m and closed 14% of the move-2 tracking gap between the robust controller and the oracle, because the unknown position of the box dominates its torque bound. COP differences cancel that position and, with the same sensor, ended at $[0.131, 0.149]$ m and closed 77%. With the box tracked the interval ended at $[0.131, 0.149]$ m and closed 81%.

5. Across the sweep, at $h = 0.08, 0.11, 0.14, 0.17$ m, COP differences closed 98, 100, 77, 0% of that gap and the tracked variant 98, 101, 81, 0%. The gain is largest when the true box is far from the conservative end of the prior, and vanishes near that end, where the robust rows are already almost exact.
6. The calibrated bounds held in every adaptive run. The residuals of the true parameters reached at most 66% of the force bound and 43% of the torque bound of the direct model, and 44% of the bound of the COP differences. The box shifted on the tray by at most 1.62 mm, inside $\bar{d} = 2.5$ mm, and its creep between paired windows used at most 65% of the creep bound.

12 Real-time implementation

Every sample is projected at every rollout step, so the cost of the controller is the cost of KH small quadratic programs and model evaluations per control cycle. The numpy reference evaluates them batched over the samples but step by step in Python, and needs about 0.2–0.3 s per plan. Porting the same operations to GPU tensor operations does not help much: with CUDA graphs the rollout still launches hundreds of small kernels per step, and it was only about seven times faster.

The implementation used here assigns one GPU thread to each sample and runs its whole rollout in one kernel (NVIDIA Warp, float32). Per step, the thread evaluates the model (forward kinematics and the mass matrix in one pass over the links, the tool Jacobian and $\dot{J}\dot{q}$), builds the rows, solves the inertia-metric projection exactly by a dual active-set method, integrates, and accumulates the cost. The weights and the raw average are computed on the GPU afterwards, and the final projection of Algorithm 1 is a second launch of the same kernel with one thread, one step and zero noise, so the executed command never leaves the device code. With the same noise, the projected inputs of every sample agree with the numpy reference to 4×10^{-5} or better, the costs to 10^{-5} , and the executed command to 5×10^{-5} . With the raw average, noise clipped at $\pm 3\sigma$ and tightened rows, the executed commands of two consecutive cycles agree to 5×10^{-5} as well, and the GPU command satisfies the tightened support rows to 2×10^{-7} . These checks are `umppi/tests/test_warp.py` and `test_warp_writing.py`; the waiter experiments of Sections 11 and 13 run on this controller.

Table 10: Plan time per control cycle in milliseconds, with the kernel time in parentheses: numpy reference at $K = 256$, and the GPU controllers on an RTX 4060 for three sample counts. Waiter: $H = 20$, control period 20 ms. Writing: $H = 15$, control period 10 ms.

Controller	numpy, $K = 256$	GPU, $K = 256$	GPU, $K = 1024$	GPU, $K = 4096$
Waiter, unified (robust rows)	293	8.68 (6.97)	8.65 (7.03)	17.30 (15.57)
Writing, unified (separable)	184	3.85 (0.72)	4.43 (1.24)	6.92 (1.95)
Writing, joint projection	200	6.46 (2.43)	7.64 (2.82)	10.89 (4.03)

Findings.

1. Both controllers run well inside their control periods: 4.4 ms of 10 ms for writing and 8.6 ms of 20 ms for the waiter at $K = 1024$, 48 and 34 times faster than the numpy reference at $K = 256$.

The plan time is nearly flat up to $K = 1024$, where one thread’s sequential rollout sets it, and grows beyond, where the device is saturated. The workstation was shared with other CPU-bound jobs during the measurement, which inflates the host-side part of the plan time (the weights, the average, the reaction map of the measured state and the launches), not the kernel time.

2. On the GPU the separable solver of Theorem 1 takes 52–70% less kernel time than the joint projection, against 8% in numpy, where the model evaluation dominates. The reaction becomes a closed-form clamp instead of two rows of the program, and the tangential program has only box rows besides the equalities.
3. The waiter kernel is slower than the writing kernel because it carries 34 rows, 17 of them support rows, and a longer horizon; it could be pruned further, but it already meets the 20 ms period with margin.

13 Baselines and the averaging ablation

Four baselines share the model, the rows that they keep, the sample count, the horizon and the clipped noise draws of Algorithm 1. A1 is penalty MPPI: the robust support rows (on the waiter) or the reaction band (in writing) become a cost $w_s \sum \max(0, \cdot)^2$, the other rows stay hard, and the average is executed. A2 is Algorithm 1 with those rows penalised instead of projected in the rollouts: the raw average is projected once onto all rows, as a safety filter would. It is run with and without the tightening, and with the tightening its penalty acts on the tightened rows as well. A3 is a one-step quadratic program in the manner of task-space inverse dynamics. It tracks a PD law in task space with the same hard rows in the metric of its objective, and has no horizon. B1 is certainty equivalence: Algorithm 1 with the point estimate of h from recursive least squares on (m, mh, md) over the same F/T windows, started from the middle of the prior or from the nominal $h = 0.07$ m. The weight w_s was swept over $\{10^3, \dots, 10^6\}$ on the waiter and over $\{1, \dots, 10^3\}$ in writing, and the gain ω of A3 over $\{1.5, 2.5, 4, 6, 9, 15\}$ rad/s. Each baseline is reported at its best value at $h = 0.14$ m, which is then kept for the other heights. The ablation runs the unified controller with the average of the earlier controllers (the projected samples) and with the raw average, each with and without the tightening $\varepsilon = 0.165$ m/s² of Section 11.

Hard rows against a penalty. Penalty MPPI is as safe as its weight. At $h = 0.14$ m it tipped the box in 5/5 with $w_s = 10^3$; 2/5 with $w_s = 10^4$; 0/5 with $w_s = 10^5$; 0/5 with $w_s = 10^6$. The best weight, 10^5 , tracked at 36.2 mm, but at $h = 0.17$ m, near the top of the prior where the robust rows are nearly the true ones, its commands exceeded the true rows (model ratio 1.09) and the box reached a COP ratio of 1.00. The robust rows of Algorithm 1 stayed at 0.80 and 0.79 there, as the certificate of Proposition 10 requires.

The average, and where the margin belongs. With the average of the earlier controllers and no tightening, the robust controller tracked at 46.8 mm and the box stayed below a COP ratio of 0.83. Averaging the raw inputs instead and projecting once, still without tightening, gave 30.2 mm, but with the command on the robust bound (model ratio 0.94 at $h = 0.17$ m) the real box reached 1.00, and the oracle with the raw average put its box on the edge (1.00). The distortion of the projected average (Proposition 6(b)) had kept the command off the binding rows, and that interior margin had absorbed the execution error without any statement about its size. Algorithm 1 moves the margin into the rows: with the raw average and the tightening it tracked at 39.2 mm, 16 % better than the earlier average, and no certified controller exceeded a true COP ratio of 0.87 at any height (the robust one 0.79). Tightening the projected average instead stacks the two margins and costs speed (62.4 mm).

Table 11: Waiter task, true $h = 0.14$ m, 5 seeds: tipped runs, tracking error of each move over the safe runs, the largest octagon ratio of the model wrench under the true parameters (above one: the rows of the true box are violated by the command), and the largest period-averaged COP ratio of the real box (one: the box is on its edge). Rows without $\varepsilon = 0$ are tightened by $\varepsilon = 0.165$ m/s².

Controller	Tipped	RMSE [mm]		Model	True
		move 1	move 2	row ratio	COP ratio
<i>Unified controller, Algorithm 1</i>					
Robust	0/5	39.2	41.2	0.66	0.63
Adaptive, COP differences	0/5	39.1	30.4	0.82	0.80
Adaptive, box tracked	0/5	28.0	29.8	0.82	0.82
Oracle	0/5	25.7	27.1	0.88	0.87
<i>Baselines</i>					
Nominal ($h = 0.07$ m)	5/5	—	—	1.87	1.00
A1 penalty MPPI ($w_s = 10^5$)	0/5	36.2	37.3	0.90	0.83
A2 penalty + filter ($w_s = 10^4$), $\varepsilon = 0$	0/5	36.3	37.7	0.78	0.77
A2 penalty + filter ($w_s = 10^5$)	0/5	48.2	46.5	0.66	0.62
A3 one-step QP ($\omega = 6$)	0/5	76.5	80.3	0.78	0.75
B1 CE, RLS from $h_0 = 0.115$	0/5	27.3	27.2	0.88	0.85
B1 CE, RLS from $h_0 = 0.07$	0/5	28.4	27.2	0.90	0.86
<i>Averaging ablation of the unified controller</i>					
Robust, projected average, $\varepsilon = 0$	0/5	46.8	47.3	0.73	0.69
Robust, projected average	0/5	62.4	57.2	0.60	0.57
Robust, raw average, $\varepsilon = 0$	0/5	30.2	31.9	0.78	0.77
Adaptive COP, projected average, $\varepsilon = 0$	0/5	45.3	36.3	0.87	0.83
Adaptive COP, raw average, $\varepsilon = 0$	0/5	30.1	24.9	0.95	0.96
Oracle, projected average, $\varepsilon = 0$	0/5	31.0	32.0	0.94	0.89
Oracle, raw average, $\varepsilon = 0$	0/5	22.5	23.6	1.00	1.00

Adaptation keeps its gain: with COP differences, move 2 took 30.4 mm against 41.2 mm for the robust controller and 36.3 mm with the earlier average. The oracle went from 31.0 to 25.7 mm, and to 22.5 mm with neither margin.

Projection inside the rollouts, at equal certificates. A2 shares everything with Algorithm 1 except that its rollouts penalise the support rows. With the same tightened rows in its cost and its filter it tracked at 48.2 mm at best ($w_s = 10^5$; 48.2–60.4 mm over the sweep), 19 % worse than Algorithm 1, because its rollouts are scored on commands that the filter then changes. Without tightening A2 lost by a similar fraction, 17 % (36.3 mm at $w_s = 10^4$ against 30.2 mm; 36.3–76.8 mm over the sweep).

The horizon. Without a horizon, A3 with the gain of the other controllers ($\omega = 15$ rad/s) accelerated at the rows’ limit, could not brake in time, overshoot the goal by up to 0.2 m and tipped the box at $t = 4.04$ s after its velocity rows saturated. Its best gain gave 76.5 mm. The reference asks for up to 3.61 m/s², more than the rows admit at any height of the sweep (at most 2.27 m/s² at $h = 0.08$ m, before tightening), so the braking must be planned. In writing, where the rows are not active on the time scale of the horizon, the same kind of controller tracked better than MPPI (Table 13).

Certainty equivalence. With the same rows and tightening, the RLS estimate came within 2.7 mm of the oracle from either prior (27.3 and 28.4 mm at $h = 0.14$ m, against 25.7 mm; move 2 at $h = 0.17$ m 38.2 against 38.1 mm) and tipped the box in 0 of 28 runs, with a model ratio of at most 0.93, because the first move excites h well before its acceleration peak. It also tracked move 2 better than the certified COP estimator (27.2 against 30.4 mm at $h = 0.14$ m), so in this task the certified estimator

Table 12: Sweep over the true height: tipped runs and the tracking error of move 2 [mm] over the safe runs; the last column gives the model row ratio and the true COP ratio at $h = 0.17$ m.

Controller	$h = 0.08$	$h = 0.11$	$h = 0.14$	$h = 0.17$	Ratios
Robust	0/3, 42.0	0/3, 42.0	0/5, 41.2	0/3, 42.0	0.80 / 0.79
Robust, projected average, $\varepsilon = 0$	0/3, 48.0	0/3, 48.0	0/5, 47.3	0/3, 48.0	0.89 / 0.83
Robust, raw average, $\varepsilon = 0$	0/3, 32.2	0/3, 32.2	0/5, 31.9	0/3, 32.2	0.94 / 1.00
Adaptive, COP differences	0/3, 21.3	0/3, 23.0	0/5, 30.4	0/3, 42.0	0.80 / 0.79
Adaptive COP, projected average, $\varepsilon = 0$	0/3, 21.5	0/3, 26.1	0/5, 36.3	0/3, 48.0	0.89 / 0.83
A1 penalty MPPI ($w_s = 10^5$)	0/3, 37.3	0/3, 37.3	0/5, 37.3	0/3, 37.3	1.09 / 1.00
A2 penalty + filter, $\varepsilon = 0$	0/3, 38.4	0/3, 38.4	0/5, 37.7	0/3, 38.4	0.94 / 0.96
A2 penalty + filter	0/3, 46.6	0/3, 46.6	0/5, 46.5	0/3, 46.6	0.80 / 0.76
A3 one-step QP ($\omega = 6$)	0/3, 80.3	0/3, 80.3	0/5, 80.3	0/3, 80.3	0.94 / 0.94
B1 CE, RLS from $h_0 = 0.07$	0/3, 20.9	0/3, 23.1	0/5, 27.2	0/3, 38.2	0.92 / 0.90

buys the guarantee at a cost in performance. A case where certainty equivalence fails, such as a biased sensor or a poorly exciting first move, remains to be shown (open item).

Table 13: Writing task, three seeds and three surface offsets $\delta \in \{-1, 0, 1\}$ mm (the order-2 controllers do not use the position of the surface, so the offsets act as repetitions): tracking error over the executed part, contact force, share of time in the band [2, 4] N, range of the commanded reaction, and runs stopped because the error passed 10 cm. The safe-MPPI rows (Section 13, last paragraph) use $\delta = 0$ and three seeds.

Controller	RMSE [mm]	Force [N]	In band [%]	λ_{cmd} [N]	Stopped
Unified, Algorithm 1 (GPU)	1.58	3.03 ± 0.45	99.3	2.12–3.39	0/9
A3 one-step QP	1.00	3.03 ± 0.42	100.0	3.00–3.00	0/9
A1 penalty MPPI ($w_s = 1$)	5.37	2.99 ± 0.65	87.5	0.62–4.17	0/9
A1 penalty MPPI ($w_s = 10$)	23.08	2.99 ± 0.65	87.8	0.74–4.13	0/9
A1 penalty MPPI ($w_s = 100$)	54.72	3.29 ± 0.48	92.3	0.73–3.90	9/9
A1 penalty MPPI ($w_s = 1000$)	67.81	3.30 ± 0.42	94.4	0.73–4.03	0/9
A2 penalty + filter ($w_s = 1$)	5.37	2.99 ± 0.65	87.5	2.00–4.00	0/9
A2 penalty + filter ($w_s = 10$)	23.08	2.99 ± 0.64	88.0	2.00–4.00	0/9
A2 penalty + filter ($w_s = 100$)	54.71	3.29 ± 0.48	92.2	2.00–3.90	9/9
A2 penalty + filter ($w_s = 1000$)	67.80	3.30 ± 0.42	94.4	2.00–4.00	0/9
Shield-MPPI ($C = 10$)	4.56	2.99 ± 0.66	87.4	2.00–3.99	0/3
Shield-MPPI ($C = 100$)	15.08	3.00 ± 0.67	86.7	2.00–4.00	0/3
Shield-MPPI ($C = 1000$)	50.00	3.27 ± 0.58	88.3	2.13–3.96	3/3
DualGuard MPPI	1.55	3.04 ± 0.42	100.0	2.89–3.12	0/3
GS-MPPI ($\rho = 5$)	1.32	2.10 ± 0.40	50.7	2.03–3.96	0/3
GS-MPPI ($\rho = 20$)	1.32	2.09 ± 0.44	47.0	2.00–4.00	0/3
GS-MPPI ($\rho = 100$)	1.32	2.09 ± 0.44	47.0	2.00–4.00	0/3

Writing. With the band as a penalty, MPPI could not both track and hold the band: from $w_s = 1$ to 10^3 the error grew from 5.4 to 67.8 mm while the force stayed in the band only 87–94 % of the time; 9 of 9 runs at $w_s = 100$ were stopped between $t = 4.4$ and 4.7 s, when the error passed 10 cm. The normal component of the sampling noise moves the reaction of every sample by newtons, and the penalty then ranks samples by that noise instead of by tracking. The clamp of the unified controller removes this noise from the costs, and the controller tracked at 1.58 mm with the force in the band 99.3 % of the time. The filter of A2 kept the commanded reaction in the band, whereas A1 commanded as little as 0.62 N, but it did not change the tracking. The one-step program A3 tracked better still (1.00 mm, 100.0 % in band). The writing task tests the treatment of the reaction and of the virtual

rows, and a horizon is not needed for it.

Safe MPPI and gradient-based MPC. Five further controllers were re-implemented on the same model, rows, costs, horizon and clipped noise draws (Table 14). Shield-MPPI [23] penalises the discrete CBF condition of the joint barriers and the violation of the support rows in its rollouts and repairs the executed control by fixed-step gradient ascent over its first three steps. DualGuard MPPI [25] passes every sample and the output through a least-restrictive filter, with the Hamilton–Jacobi value of a braking double integrator for each joint limit and zero tangential tray acceleration as the safe input for the support rows, and averages the filtered samples. GS-MPPI [24] composes all rows by a soft-min and applies its closed-form safe law to every sample and to the raw mean; because the support rows constrain the input directly, the composite is not affine in the input and the law is repeated until it holds. None of the three handles equalities, so all keep the tray orientation by the equality projection of Algorithm 1, and each was tuned at $h = 0.14$ m like the other baselines (Shield: penalty weight, GS: soft-min parameter). The gradient-based controllers follow Heins and Schoellig [39]: an offline robust trajectory optimisation over the whole run with the vertex rows softened by an L2 slack, tracked by PD feedback, and a receding-horizon robust NMPC with the cost, horizon and step of the MPPI controllers, the rows hard at the first knot and a jerk penalty.

Table 14: Safe MPPI and gradient-based MPC on the waiter, true $h = 0.14$ m (5 seeds for the MPPI controllers; the gradient-based ones are deterministic): tipped runs, tracking error, the model row ratio and true COP ratio as in Table 11, and the largest period-averaged execution error of the tray acceleration. GS-MPPI applies its safe law until the composite holds (at most five times), the one-step variant once, as published. Upper block: rows as they are; lower block: rows tightened by $\varepsilon = 0.165$ m/s².

Controller	Tipped	RMSE [mm]		Model ratio	True COP	Exec. err. [m/s ²]
		move 1	move 2			
Algorithm 1	0/5	30.2	31.9	0.78	0.77	0.079
Shield-MPPI ($C = 1000$)	0/5	40.5	40.2	0.78	0.72	0.065
DualGuard MPPI	0/5	85.4	114.1	0.60	0.59	0.204
GS-MPPI ($\rho = 100$)	0/5	30.2	32.1	0.78	0.79	0.078
GS-MPPI, one step ($\rho = 20$)	0/5	28.8	31.2	1.08	1.00	0.075
Offline robust plan	0/1	23.1	22.2	1.02	0.96	0.108
Online robust NMPC ($w_j = 0.1$)	0/1	34.5	37.1	0.78	0.74	0.060
Algorithm 1	0/5	39.2	41.2	0.66	0.63	0.063
Shield-MPPI ($C = 1000$)	0/5	53.6	49.5	0.66	0.61	0.055
DualGuard MPPI	0/5	110.2	237.7	0.52	0.48	0.257
GS-MPPI ($\rho = 20$)	0/5	39.3	41.4	0.66	0.63	0.059
GS-MPPI, one step ($\rho = 20$)	0/5	34.4	37.3	1.00	0.92	0.072
Offline robust plan	0/1	27.7	26.7	0.85	0.83	0.109
Online robust NMPC ($w_j = 0.1$)	0/1	45.1	46.1	0.65	0.64	0.052

On the waiter, GS-MPPI matched Algorithm 1: 30.2 and 39.3 mm on move 1 with the rows as they are and tightened, against 30.2 and 39.2 mm. With one binding row at a time its soft-min composite reduces to that row and its safe law to a projection onto it, and it too averages the raw inputs and filters the output, so without physical equalities the two schemes coincide in structure; its closed-form filter is also cheaper than an exact projection. They differ where several rows bind at once, which the scenario at the end of this section tests. GS-MPPI composes barrier functions of the state, for which the composite CBF condition is affine in the input and one application of its safe law is exact. The support rows constrain the input directly, so their soft-min is concave in the input, and one application lands on the zero of its linearisation, where the composite is still negative; the GS-MPPI

controller above therefore applies the law until the composite holds, at most five times. Applied once, as published, the law left the executed command outside the true rows: with $\rho = 20$ the model ratio reached 1.08 at $h = 0.14$ m (1.12 with $\rho = 100$) and 1.23 at $h = 0.17$ m, where 1 of 3 runs tipped, and with the tightened rows it still reached 1.18 at $h = 0.17$ m, with the box on its edge (true COP ratio 1.00). The smaller tracking error of this variant (28.8 and 34.4 mm on move 1) is bought by the violation. Shield-MPPI kept the rows through its repair but tracked at 40.5 and 53.6 mm, because its rollouts are scored on commands that the repair then changes, and the switch of DualGuard MPPI to the most conservative input tracked at 85.4 and 110.2 mm. Over the four heights, with the rows as they are, Shield-MPPI had 0 tips, a model ratio of at most 0.94 and a true COP ratio of at most 0.89; DualGuard MPPI had 0 tips, a model ratio of at most 0.73 and a true COP ratio of at most 0.74; GS-MPPI had 0 tips, a model ratio of at most 0.94 and a true COP ratio of at most 1.00. GS-MPPI put the box on its edge at $h = 0.17$ m (true COP ratio 1.00), as Algorithm 1 does without the margin, and with the tightened rows it stayed at 0.79: any scheme that rides the boundary of the rows needs the calibrated margin. The offline plan tracked best (23.1 and 27.7 mm), since it optimises the whole run with the reference known in advance. It certifies the plan, not the executed command: the tracking feedback pushed the model ratio to 1.02 at $h = 0.14$ m and to 1.24 at $h = 0.17$ m, where the box tipped, and with tightened rows the box still reached a COP ratio of 0.98 there. The online NMPC, with the cost, horizon and step of the MPPI controllers, kept the rows and tracked at 34.5 and 45.1 mm, 14 and 15 % behind Algorithm 1, and IPOPT needed 451 ms per step, against 9 ms for the GPU controller of Algorithm 1 at the same sample count.

On the writing task (Table 13), with the contact handled by the Gauss dynamics of Algorithm 1 for all of them, Shield-MPPI held the band 87–88 % of the time and tracked at 4.6–50.0 mm over its parameter; DualGuard MPPI held the band 100 % of the time and tracked at 1.6 mm; GS-MPPI held the band 47–51 % of the time and tracked at 1.3 mm over its parameter; Algorithm 1 held it 99 % of the time at 1.6 mm. No baseline did well on both tasks. GS-MPPI matched Algorithm 1 on the waiter but lost the band in writing, where its raw average passes the exploration noise along the contact normal to the reaction. DualGuard MPPI held the band in writing, where its average of filtered samples acts like the clamped average of Algorithm 1, but was the most conservative controller on the waiter. Algorithm 1 takes the raw average for the motion and the clamped average for the reaction, which separation allows.

On the dual-arm task (rows as they are, $\delta_b = 0$, three seeds), Shield-MPPI kept the joint-4 bound (goal error 0.11 rad, internal force 1.41 N); DualGuard MPPI kept the joint-4 bound (goal error 0.11 rad, internal force 1.52 N); GS-MPPI crossed the joint-4 bound by 0.002 rad (goal error 0.07 rad, internal force 1.09 N); Algorithm 1 crossed it by 0.008 rad without the margin and kept it with the calibrated one (goal error 0.26 rad). Shield-MPPI and DualGuard MPPI kept the bound closer to the goal than Algorithm 1: a barrier on the joint position absorbs the quasi-static sag of the unmodelled tray directly, whereas the acceleration bound of Algorithm 1 is set by the oscillation of the grasp springs and is conservative here; a position-level execution bound, or the tray load in the model, would remove the difference (open item).

Several binding rows. In the waiter runs above, the moves run along a face normal of the octagons, and one support row binds at a time. To load a vertex of the rows, the box was given a low friction coefficient, true $\mu = 0.12$ with the prior $[0.10, 0.15]$, so that the slip rows bind before the tipping rows, and the moves were turned to 67.5° from the tray axes, the direction of a vertex of both octagons of support rows, and shortened to 0.6 s. The execution-error bound was calibrated for this scenario as in Section 11, $\varepsilon = 0.337$ m/s², and Table 15 lists the controllers with the robust prior at $h = 0.14$ m, with the parameters tuned on the waiter above.

No controller tipped the box. Among the controllers with the robust prior that kept the rows,

Table 15: Several binding rows: the waiter with a low-friction box (true $\mu = 0.12$, prior $[0.10, 0.15]$) moved at 67.5° to the tray axes in 0.6s; true $h = 0.14$ m, robust prior, 5 seeds. Tipping and slip ratio: the largest ratio of the executed command to the true tipping and slip rows; slide: the largest displacement of the box on the tray. Upper block: rows as they are; lower block: rows tightened by $\varepsilon = 0.337$ m/s².

Controller	Tipped	RMSE [mm]		Tipping ratio	Slip ratio	True COP	Slide [mm]
		move 1	move 2				
Algorithm 1	0/5	49.9	55.0	0.70	0.83	0.70	1.4
Algorithm 1, projected average	0/5	73.7	77.0	0.65	0.77	0.62	1.4
GS-MPPI ($\rho = 20$)	0/5	52.2	56.7	0.70	0.83	0.69	1.3
GS-MPPI ($\rho = 100$)	0/5	51.5	56.2	0.70	0.83	0.69	1.3
GS-MPPI ($\rho = 1000$)	0/5	51.5	56.3	0.70	0.83	0.69	1.3
GS-MPPI, one step ($\rho = 20$)	0/5	50.1	54.6	0.89	1.06	0.81	1.4
Shield-MPPI ($C = 1000$)	0/5	70.5	85.5	0.70	0.83	0.66	1.4
DualGuard MPPI	0/5	122.4	364.1	0.50	0.59	0.45	1.1
Oracle (true h, μ)	0/5	39.7	45.4	0.84	1.00	0.87	4.5
Algorithm 1	0/5	84.0	82.2	0.47	0.56	0.44	1.2
Algorithm 1, projected average	0/5	125.9	376.8	0.43	0.51	0.53	2.9
GS-MPPI ($\rho = 20$)	0/5	87.8	84.8	0.46	0.55	0.44	1.2
GS-MPPI ($\rho = 100$)	0/5	86.2	84.9	0.46	0.55	0.43	1.1
GS-MPPI ($\rho = 1000$)	0/5	86.3	85.5	0.46	0.55	0.43	1.1
GS-MPPI, one step ($\rho = 20$)	0/5	75.0	74.5	2.17	2.59	0.84	2.9
Shield-MPPI ($C = 1000$)	0/5	113.2	228.2	0.46	0.55	0.43	1.3
DualGuard MPPI	0/5	186.4	519.2	0.38	0.45	0.44	3.3
Oracle (true h, μ)	0/5	62.6	67.5	0.60	0.72	0.57	1.3

Algorithm 1 tracked best, at 49.9 and 84.0 mm on move 1 with the rows as they are and tightened. GS-MPPI, with its law applied until the composite holds, followed 3–5 % behind in every seed, and neither a sharper soft-min ($\rho = 1000$) nor twenty applications of the law closed the gap (51.5 and 86.3 mm). In one run (seed 0, tightened rows) the raw mean violated the rows in 163 plans, and in 55 of them its projection landed on a vertex, with two rows active. There the law, applied until the composite held with $\rho = 1000$, ended on one of the two rows and inside the other, by 0.08 rad/s^2 in the median and up to 0.14 rad/s^2 in the distance of the command to that row, and it moved the command 8 % farther from the raw mean in the inertia metric than the projection did. A step onto one row also moves the command to the feasible side of the adjacent row, so a sequence of such steps stops on a face; the projection treats the active rows together and lands on the vertex. Applied once, the law left the command outside the true rows, with slip ratios of up to 1.06 and 2.59 and tipping ratios of up to 0.89 and 2.17. In its run of seed 0 with the tightened rows, up to 12 rows were violated at the raw mean, the soft-min was dominated by the most violated row, and after one application rows were still violated in 80 of 195 plans, up to 8 of them; after two applications at most 2 and after five none. The box slid by up to 2.9 mm, against at most 1.4 mm under Algorithm 1 and GS-MPPI applied to convergence. Shield-MPPI kept the rows but tracked at 70.5 and 113.2 mm on move 1 and 85.5 and 228.2 mm on move 2; with the rows as they are it never came within 1 cm of the first target, and with the tightened rows it never came within 1 cm of the first or the second target. DualGuard MPPI tracked at 122.4 and 186.4 mm on move 1 and never came within 1 cm of the second target. Its safe input for the support rows, zero tangential tray acceleration, keeps the tray velocity, so a braking sample that violates the rows is replaced by one that does not brake: in the run of seed 0 with the rows as they are, its filter replaced 20 % of the samples on average and up to 47 %, and the tray passed the first target by 111 mm. With the projection as its safe input, DualGuard MPPI would compute, up to its joint-limit filter, the projected average of the ablation, which tracked at 73.7 and 125.9 mm; the

raw average of Algorithm 1 again did better by 32 and 33 %. With the true parameters Algorithm 1 tracked at 39.7 and 62.6 mm, and without the margin its command rode the true slip rows (ratio 1.00) and the box slid by up to 4.5 mm; the difference in tracking is the price of the lower friction bound of the prior, which caps the slip ratio at 0.83.

14 Numerical verification

All checks run in about two minutes with `python run_all.py` inside `tier2/verify`, which writes the log `run_all.log`. QPs are solved exactly by a Lawson–Hanson least-distance program, cross-checked against Clarabel to 10^{-8} .

Table 16: Verification results (54 checks, all passing).

Script	Check	Result
v1	Identity (3), 200 random systems	3.7×10^{-15}
v1	Joint M -projection equals the two separated QPs	2×10^{-12}
v1	Leakage under $W = M / M + J_c^T S J_c / I$ / random	$< 10^{-11} / < 10^{-11} / 83 / 330$
v1	Reaction rows leave the motion LP optimum unchanged	3×10^{-14}
v1	Torque limits: separated answer infeasible / sequential breaks rows	193/200, 169/200
v1	Passive coordinate: band bounds the motion LP	50/50
v2	Eq. (5), 4 loop/environment pairs, 0.05–2 mm	rel. err. $\leq 2 \times 10^{-2}$
v2	Order-2 commanded reaction is offset independent	10^{-15}
v3	Order 1: layer slope, closed form, per-step gains	1.00, 0.998–1.000, 2.00
v3	Order 2 convex: layer slope, closed form, per-step gains	1.01, 1.00–1.02, 2.01
v3	Order 2 concave: favourable sign	margin > 0
v3	Sphere, ZOH physics: fixed / per-step gains	slopes 1.05 / 2.97
v3	Sphere, ZOH: layer against $\Delta t \ddot{h} / (2k_1 k_2)$	ratio 1.03–1.06
v4	Fixed H , orders 1 / 2 (traj/cost/update)	2.02/2.00/2.23, 2.10/2.14/3.04
v4	Fixed T_h , orders 1 / 2 (traj/cost/update)	0.96/1.00/0.89, 0.92/0.95/0.95
v4	White-noise exploration: Itô drift	0.0365 vs 0.036
v5	Feasibility and fixed point at $t = 0$	4×10^{-14}
v5	Identity (9) / bound ratio (max, median)	$8 \times 10^{-14} / 0.33, 0.10$
v5	Same face / $\beta \rightarrow 0$	$7 \times 10^{-12} / 0$
v6	Projected SNIS: MSE/bound, bias/bound (bound $(4\rho - 2)/K$), MSE slope in K	0.07, ≤ 0.10 , -1.00
v7	Deadbeat virtual rows: steady residual slope / fixed gains monotone	2.09 / yes
arm	Controller model against MuJoCo ($M, b, J, \dot{J}\dot{q}$)	2×10^{-15}
arm	Thm. 2(i)–(iii): band, motion rows, virtual rows, separable vs joint	0, 0, 3×10^{-15} , 9×10^{-15}
v8	Prop. 9: box maximum of 300 multilinear rows at a vertex / quadratic control	≤ 0 / interior peak
v8	Waiter rows affine in (h, μ) ; executed command inside the rows of 35 box parameters	4×10^{-15} ; 2×10^{-16}
v8	Prop. 12: containment and nesting, 24 000 windows / noise at 3 times the bound	yes / broken
v8	Prop. 11: tightened rows under the worst deviation / untightened control	1×10^{-15} / violated
v8	COP differences, creep and noise inside the bounds, 30 000 windows / 3 times the bounds	contained, nested / broken

15 Corrections to the earlier plan

The earlier plan labelled the CAMP tray task as coupled. It has no manifold, and its support-wrench rows are affine in $\ddot{q}$, so they are motion rows (Section 4). Passive bodies inside the constrained system

and torque limits are two sources of coupling (Proposition 1); an experiment on coupled rows must include such a source rather than identify the waiter support rows as coupled.

The earlier plan proposed a wrench-bounded retraction at order 2. For a rigid physical constraint the normal motion is fixed by the environment, so there is nothing for the controller to restore, and restoring towards a mis-modelled manifold is harmful (Proposition 2). The retraction survives for virtual equalities only.

The June Proposition is correct at fixed horizon length. With its unweighted running cost it is $O(T_h^2)$ at fixed horizon time and does not vanish as $\Delta t \rightarrow 0$. The fixed-horizon-time statement of Proposition 5, with the Δt -weighted cost, is the one that vanishes, and its bounded-input assumption excludes white-noise-scaled exploration.

The earlier plan, like both earlier controllers, executed the average of the projected samples, admissible by convexity. The baselines of Section 13 showed that this average stays off the binding rows by the distortion of Proposition 6(b), which cost tracking and silently absorbed the execution error. Algorithm 1 now averages the raw inputs, projects the motion part of the executed command once and states the margin as a tightening of the rows. The reaction keeps the average of the clamped samples, because averaging the raw reaction as well broke force regulation in writing (Proposition 6, remark), and the convexity route remains as Proposition 3(b) and as the ablation.

16 Open items

1. State the Lipschitz constant of Π_{act} under a uniform LICQ explicitly (currently cited).
2. Extend Lemma 1 to zero-order-hold execution with explicit constants; until then the executed-motion margin of Proposition 4(c) is an estimate.
3. Run the additional log analyses listed in `PENDING.md`; these are future analyses and are not evidence for the present results.
4. Design the torque-level bimanual friction task so that coupled rows are active (Proposition 1).
5. Reduce the tray tilt of the order-2 controller on the dual-arm task by modelling the tray load (Section 10); the measured-load compensation of Section 11 is one way.
6. Extend the adaptation to friction: the slip rows never bind in the waiter task, so μ stays at its prior.
7. Replace the calibrated error bounds of Section 11 by bounds derived from the contact compliance, which would turn the calibrated assumption into a proved one.
8. Schedule the execution-error bound with the admissible acceleration, $\varepsilon(h_{\text{hi}})$, instead of taking the most aggressive calibration run.
9. Bound the execution error of the joint rows at the position level: on the dual-arm task the acceleration bound is set by the oscillation of the grasp springs and keeps the arm 0.26 rad from the goal, while a barrier on the joint position (Shield-MPPI) keeps the bound at 0.11 rad (Section 13).
10. Show a case where certainty equivalence fails (a biased F/T offset, a poorly exciting first move), or present the value of the certified estimator as the guarantee alone.
11. Add a writing variant whose rows are active on the time scale of the horizon (a faster path, joint-limit or obstacle rows): on the current path a one-step program tracks better than MPPI.

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